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METABOLISM OF RAINBOW TROUT, SALMO
GAIRDNERI.**

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FACTORS INFLUENCING RESPIRATORY METABOLISM
OF RAINBOW TROUT, SALMO GAIKDNERI

by

Ian William Dickson

A dissertation submitted in partial fulfillment
of the requirements for the degree

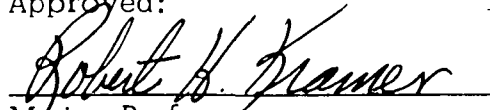
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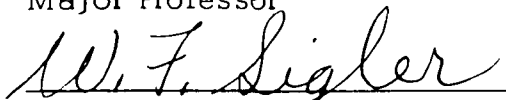
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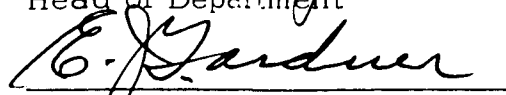
in

Fishery Biology

Approved:


Major Professor


Head of Department


Dean of Graduate Studies

UTAH STATE UNIVERSITY
Logan, Utah

1968

He paused and stood for a moment in silence, drumming with his fingers on the glass of the aquarium. Poised between mud and air, the two obese and aged carp hung in their greenish twilight, serenely unaware of him. Dr. Obispo shook his head at them. The worst experimental animals in the world, he said in a tone of resentment mingled with a certain gloomy pride. Nobody had a right to talk about technical difficulties who hadn't tried to work with fish. Take the simplest operation; it was a nightmare. Had you ever tried to keep its gills properly wet while it was anaesthetized on the operating-table? Or, alternatively, to do your surgery under water? Had you ever set out to determine a fish's basal metabolism, or take an electro-cardiograph of its heart action, or measure its blood pressure? Had you ever wanted to analyze its excreta? And, if so, did you know how hard it was even to collect them? Had you ever attempted to study the chemistry of a fish's digestion and assimilation? To determine its blood pressure under different conditions? To measure the speed of its nervous reactions?

No, you had not, said Dr. Obispo contemptuously. And until you had, you had no right to complain about anything.

Aldous Huxley,

After Many A Summer

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ABSTRACT

Factors Influencing Respiratory Metabolism of

Rainbow Trout, Salmo gairdneri

by

Ian William Dickson

Utah State University, 1968

Major Professor: Dr. R. H. Kramer

Department: Wildlife Resources

The effect of temperature, starvation, and time of day on scope for activity, and the influence of sex and season on active metabolism of rainbow trout were studied from October, 1966 to December, 1967. Four hundred and twenty-two active and 196 standard metabolism measurements were made on hatchery and wild trout strains acclimated to 5, 10, 15, 20, or 25 C.

Standard metabolism of hatchery and wild trout acclimated to 5, 10, 15, 20, and 25 C increased significantly with increased temperature. With the exception of 15 C, standard metabolic rates of the two strains were not significantly different at the test temperatures. Active metabolism and scope for activity were maximum at 15 and 20 C for hatchery and wild trout, respectively, and were decreased at 25 C. Active metabolism and scope for activity of wild trout were significantly higher than these rates for hatchery trout. At temperatures up to 20 C, active metabolic rates were about 9-11 times higher than the standard rates.

Active metabolism of hatchery trout acclimated to 15 C was not significantly affected by 1 to 9 days starvation. Standard metabolism was decreased significantly following 2 days starvation. Maximum scope for activity was observed at 6 days starvation.

Time of day (forenoon vs afternoon) had no significant effect on standard and active metabolism, or on scope for activity of wild trout acclimated to 15 C and subjected to continuous light and controlled periods of light and dark. Also, light conditions had no significant effect on active and standard metabolic rates.

Seasonal trends in active metabolic rates were found for both sexes; highest active rates of oxygen consumption occurred during the trout spawning period.

Active rates of oxygen consumption of males were consistently higher than those of females. Analysis of covariance showed these differences to be significant at the 1 and 6 per cent levels.

(118 pages)

INTRODUCTION

Fish respiratory metabolism studies have generally involved measurements of oxygen consumption, and the metabolic rate of fish has been usually reported as the amount of oxygen consumed by a given weight of fish in a specified time. Respiratory metabolism of fish has been reported as standard, routine, and active (Fry, 1957), each in relation to activity.

Standard metabolism is the minimum rate of oxygen consumption in the absence of spontaneous activity. Fry (1967) stated that standard metabolism is obtained preferably by determining some relationship between oxygen consumption and activity and extrapolating this relation to zero activity (Spoor, 1946; Ruhland and Heusner, 1959; Blazka, Volf, and Cepela, 1960; Wohlschlag, 1962; Beamish and Mookherjee, 1964; Brett, 1964; Smit, 1965; Muir, Nelson, and Bridges, 1965). Others (Job, 1955; Schultze, 1965) have used the lowest rate of oxygen consumption in the daily cycle to approximate standard oxygen consumption. Anesthetics have been used to measure standard metabolism (McFarland, 1959) but this technique has received little attention.

Routine metabolism is the rate of oxygen consumed by fish whose only movements are spontaneous. Until recently, this was the most commonly measured rate of oxygen consumption.

Active metabolism was defined by Fry (1967) as the maximum steady rate of oxygen consumption of fish under continuous forced activity. To

achieve precision in measurements of active metabolism, it is necessary to force the fish to respire at its maximum rate.

Measurements of active and standard metabolism permit description of "scope for activity," which Fry (1947, p. 57) defined as the difference between active and standard rates of metabolism. The significance of the difference between active and standard metabolic rates is its expression of the respiration available for activity (Fry, 1957). Brett (1958) considered scope for activity as a tool for assessment of environmental stress on fishes. He suggested that any environmental factors that decrease scope may be considered as stress factors.

Scope for activity has received increased attention as a parameter for determining the "energy available" to fish. Brett (1964) used this parameter to determine the energy required by sockeye salmon swimming at various speeds. Most recently, Beamish and Dickie (1967, p. 235) indicated its significance to the study of natural production. Following a lengthy and comprehensive review of metabolism and biological production in fish, they stated, "This index of scope might justifiably be considered as a measure of the scope for total energy expenditure, irrespective of the functions for which energy is used." They speculated further that "...Measures of the metabolic relation among species may give a rough measure of the relative abilities of species to make use of their environment for work."

Scope for activity is a function of all the variables controlling active and standard metabolism. These variables may be environmental

factors such as temperatures, photoperiod, oxygen and carbon dioxide content of the water, or inherent factors such as sex, sexual development, body weight, body fat content, etc. Fry (1957) presented a comprehensive account of the literature dealing with the effects of such variables on fish respiration and metabolism. Effects of sex, season, time of day, and starvation on scope for activity have received relatively little attention.

Few studies have been made on rainbow trout respiratory metabolism, although considerable information is available concerning metabolism of brook trout, *Salvelinus fontinalis*, and salmon, *Oncorhynchus* sp. Recently, Kutty (1966) used rainbow trout, *Salmo gairdneri*, in studies of respiratory quotients of fish. Differences in behavior, growth, and survival have been reported between hatchery and wild trout strains (Flick and Webster, 1964; Smith, 1957; Vincent, 1960). Though these studies suggest that physiological differences exist among trout strains, respiration rates of wild and hatchery trout have not been compared.

Objectives

The objectives of the present study were:

1. To determine the effect of temperature on scope for activity of a wild and a hatchery strain of catchable-size rainbow trout.
2. To determine the effect of starvation for 1, 2, 3, 6, and 9 days on scope for activity of catchable-size rainbow trout acclimated to the temperature at which scope for activity was maximal as

3. To determine the effect of time of day (daylight hours) on scope for activity of a catchable-size rainbow trout strain subjected to controlled periods of light and dark or continuous lighting, and acclimated to the temperature for maximum scope as determined in objective 1.
4. To determine the effects of sex and season on respiration rate of rainbow trout acclimated to the temperature for maximum scope as determined in objective 1.

REVIEW OF LITERATURE

Apparatus

Active metabolic rates of fish have been estimated by numerous investigators employing several techniques. Fry (1947) and Fry and Hart (1948) first used an annular chamber which, when rotated, forced fish to swim against a current. Others have used similar apparatus (Graham, 1949; Gibson and Fry, 1954; Job, 1955; Wohlschlag, 1957; Wohlschlag and Juliano, 1959). Basu (1959) used an annular chamber but induced fish to swim against a current by an electrical stimulus. Blazka, Volf, and Cepela (1960) described an efficient active respirometer in which fish placed in a cylinder were forced to swim against a water current produced by a rotating propeller. A mild electric stimulus encouraged fish to swim. The Blazka-type respirometer has been used with good success by Smit (1965) and Kutty (1966). Recently, Brett (1964) used a respirometer in which water was circulated by means of a centrifugal pump. Fish were forced to swim against a current of relatively flat velocity profile characterized by micro-turbulent flow. This, and similar apparatus described by Davis et al. (1963); Katz, Pritchard, and Warren (1959); and Dahlberg, Shumway, and Doudoroff (1968) have been used in studies of swimming speed and respiration where characteristics of the velocity profile in the chamber are important.

Estimates of standard metabolic rates were obtained in studies

prior to 1946 by encouraging quiescence of fish within a chamber and using the lowest rate of oxygen consumption over an extended period, usually 24 hours, as the standard rate. However, because of spontaneous activity associated with certain fish species, e.g. goldfish, Beamish and Dickie (1967) stated that this technique has been used in studies of species for which it is unsuited. It is customary in present-day studies of standard metabolism to consider spontaneous activity when estimating standard rates of oxygen consumption. Spoor (1946) first measured spontaneous activity by equating it to the number of deflections of an aluminum paddle activated by water currents set in motion by the movements of a fish. Corti and Weber (1948) measured activity by means of a delicately suspended experimental tank which moved when the fish was active. Fontaine (1956) determined activity by equating it to deflections of a glass plate suspended in a respiration chamber caused by fish movement. Smit (1965) used photocells to record activity. McKenzie (1960) and Beamish and Mookherjee (1964) determined spontaneous activity by relating it to heat loss from a thermostatic heater probe set at a temperature slightly higher than that of the respirometer water. Fish movement generated water currents which cooled the probe and increased the time the heater was in circuit. The time the heater was in circuit was equated to spontaneous activity. This technique has been perfected further by Mathur and Shrivastava (1967).

Ruhland and Heusner (1959) employed a manometric method for measuring spontaneous activity in which swimming of fish caused waves

on the water surface which were recorded and equated to activity. Muir, Nelson, and Bridges (1965) detected fish movements in an annular chamber by means of two high frequency "Fish Finders." Movement of a fish in the transmitter field shifted the audio frequency of a reflected signal.

Brett (1964) obtained estimates of standard metabolism by measuring oxygen consumption of fish forced to swim at different speeds and extrapolating this relationship between oxygen consumption and swimming speed to zero swimming speed.

Related Studies

The influence of temperature on oxygen consumption of fish of similar size or corrected to a constant weight has been well investigated at both standard and active levels. Rarely, however, have studies been conducted to describe active and standard metabolism simultaneously for a single species at temperatures between and closely approaching the lower and upper incipient lethal temperatures; the study by Brett (1964) being a notable exception. Using sockeye salmon, *Oncorhynchus nerka*, he observed the typical response of respiratory metabolism of fish to increased temperature; standard metabolism continued to increase with increased temperature, while active metabolism increased to 15 C but decreased with further increase in temperature. Scope for activity increased with temperature to a maximum at 15 C, but decreased with increased temperature. Job (1955) found that active and standard metabolism of brook trout increased with temperature to 20 C, the highest temperature he used.

Basu (1959) determined active rates of oxygen consumption for brook trout; brown bullhead, Ameiurus nebulosus; carp, Cyprinus carpio; and the goldfish, Carassius auratus. By obtaining estimates of the standard rates for these species from the literature, he described the influence of temperature on scope for activity. However, for any one species, scope for activity was reported at no more than three acclimation temperatures.

Studies to determine differences in respiratory metabolism of different fish strains of the same species have not been reported. Job (1955) determined the effects of fish size, temperature, and partial pressure of oxygen on oxygen consumption of hatchery and wild stocks of brook trout. He noticed no difference in maximum swimming speed between the two strains and, based on these and similar observations reported for chinook salmon, Oncorhynchus tshawytscha, by Kerr (1953), Job chose to treat these fish as one stock. His data were presented in a manner such as to prevent comparison of rates of oxygen consumption for the separate trout stocks.

With the exception of Beamish (1964a), the effect of starvation on oxygen consumption has not been determined when quantitative account is taken of spontaneous activity. Beamish found that standard and routine oxygen consumption of brook trout and white suckers, Catostomus commersoni, decreased to a minimum within 2 to 3 days, after which it remained constant. Similar findings have been reported by Wells (1935) for opaleye, Girella nigricans; and by Moss and Scott (1961) for bluegill, Lepomis macrochirus; largemouth bass, Micropterus salmoides; and

channel catfish, Ictalurus punctatus.

In contrast to these findings, other studies of starved poikilotherms have shown a gradual decrease in oxygen consumption (presumably low routine) with starvation. Such findings have been reported by Smith (1935a, 1935b) who found a logarithmic decrease in oxygen consumption for starved lung-fish, Protopterus aethiopicus. Similar observations were made by Mann (1958) on an eel, Anguilla japonica; Hickman (1959) on starry flounder, Platichthys stellatus; and Berg, Lumbye, and Ockelmann (1958) on a limpet, Anacylus fluviatilis. Beamish (1964a) suggested that the rapid decrease in standard oxygen consumption of starved fish can be attributed to the reduction in oxygen requirements for the assimilation of food, whereas the gradual decrease so often reported in the literature reflects additionally a gradual subsidence of spontaneous activity. Muir, Nelson, and Bridges (1965) observed a rapid decrease in oxygen consumption by the euryhaline aholehole, Kuhlia sandvicensis, maintained at a swimming speed of 0.23 body lengths per second for 5 days. Oxygen consumption reached a maximum in about 3 days.

The effect of starvation on active metabolism has received little emphasis. Barrett (in Fry, 1957) found that the active rate of oxygen consumption of rainbow trout was unaffected by prolonged starvation.

Diurnal fluctuations in metabolism of fishes have been given considerable attention by Winberg (1956). Diurnal fluctuations in oxygen consumption have been reported in the past, but without rigorous control or measurement of activity. At present, there is no general agreement

among findings concerning diurnal effects on respiratory metabolism. Clausen (1933, 1936) reported daily fluctuations in oxygen consumption of some fresh-water species but not in others. Higginbotham (1947) showed that channel catfish had distinctly higher rates of oxygen uptake in the afternoon than in the morning. Spencer (1939) found no evidence in support of diurnal rhythm in the rate of oxygen consumption in the bluegill. Graham (1949) reported that brook trout had a daily fluctuation in standard oxygen consumption with the low point occurring between midnight and early morning. Job (1955), using brook trout, found little evidence to support Graham's conclusions. Moss and Scott (1961) found no evidence of consistent diurnal rhythms in oxygen consumption of three species of fresh-water fish. They suggested that discrepancies in the literature may well be accounted for by the lack of acceptable methods for detecting daily fluctuations of low magnitude, or possibly, restrictions on fish activity suppress any diurnal activity rhythms which would otherwise be reflected in the metabolic rates. Studies of diurnal rhythms in active rates of oxygen consumption have not been reported.

There is little information in the literature to indicate that sex has an appreciable effect on fish metabolism, although much of the work has involved sexually immature fish. Wohlschlag and Juliano (1959) found that metabolic rates (apparently routine) of females were higher than those of males at different temperatures and activities, but the differences were not statistically significant. Beamish (1964b) found that standard rates of female brook trout approximated those for males.

Seasonal differences in low routine or standard oxygen consumption have been demonstrated by Wells (1935), Privolnev (1948), and Beamish (1964b). Highest oxygen consumption of resting fish have been found to coincide with the spawning season. Seasonal changes in oxygen consumption were reported to be related to several factors of which temperature, photoperiod, and thyroid activity have received most consideration (Beamish and Dickie, 1967). No reports on studies of seasonal fluctuations in active metabolism were found in the literature.

MATERIALS AND METHODS

Fish Strains, Care, and Preparation

Wild (De Smet) and hatchery (Hull-Erickson) trout strains were raised in Utah State Fish and Game Hatcheries at Glenwood and Logan, respectively. The wild trout stock was obtained from De Smet Lake, Wyoming, and hatchery trout were produced from brood stock maintained in Utah hatcheries. De Smet trout have been exposed to natural selection processes in De Smet Lake since as early as 1912, while Hull-Erickson trout have experienced hatchery selection since 1880 (Dollar and Katz, 1964). As pointed out by Flick and Webster (1964), the term "strain" generally implies genetic overtones, and it is used in this sense here. Definite genetic differences are not implied, but differences observed in behavior, growth, and survival seem to warrant the designation.

Wild trout were allowed at least a 1-month period of adjustment to the Logan Hatchery conditions before being used in experiments. Fish maintained at the Hatchery were fed twice daily on a commercial pellet composed of: crude protein--not less than 38 per cent; crude fat--not less than 5 per cent; crude fiber--not more than 7 per cent; ash--15 per cent; and various vitamins and minerals. Fish at the Hatchery were held outdoors and were exposed to natural photoperiods. Standard experimental hatchery procedures for maintenance, prophylactic treatment, and

care of fish assured healthy specimens. Fish weighed approximately 100 to 400 g and were 10 to 14 months old.

Acclimation

All experiments were done in the Utah State University Fisheries Laboratory. Fish were acclimated to 5, 10, 15, 20, or 25 C in five insulated 400-liter tanks. Twenty-five fish were placed in each tank. Individual tanks were equipped with a temperature recorder, motor-driven aerator, fiberglass-wool filter, and thermoregulator-controlled heater (Figure 1).

Dechlorinated laboratory water was 8-10 C and was heated to 15, 20, and 25 C temperatures. Water cooled to 4 C in a Min-O-Cool (a refrigeration unit and fiberglass reservoir) was heated to 5 and 10 C. Acclimation temperatures were controlled to within ± 0.5 C.

Dissolved oxygen concentrations at all temperatures exceeded 90 per cent air saturation. This high level of dissolved oxygen was used throughout this study to denote "air saturation." At the Logan elevation of 1372 m, air saturation values for oxygen are: 10.7 ppm at 5 C; 9.5 ppm at 10 C; 8.5 ppm at 15 C; 7.7 ppm at 20 C; and 7.0 ppm at 25 C.

Water temperature in the acclimation tanks was the same as hatchery water temperature (17 C) when the fish arrived. Procedures described by Beamish and Mookherjee (1964) were followed for acclimating fish to different temperatures. Temperatures in the tanks were increased or decreased 1 C per day until the desired temperature was obtained. Fish

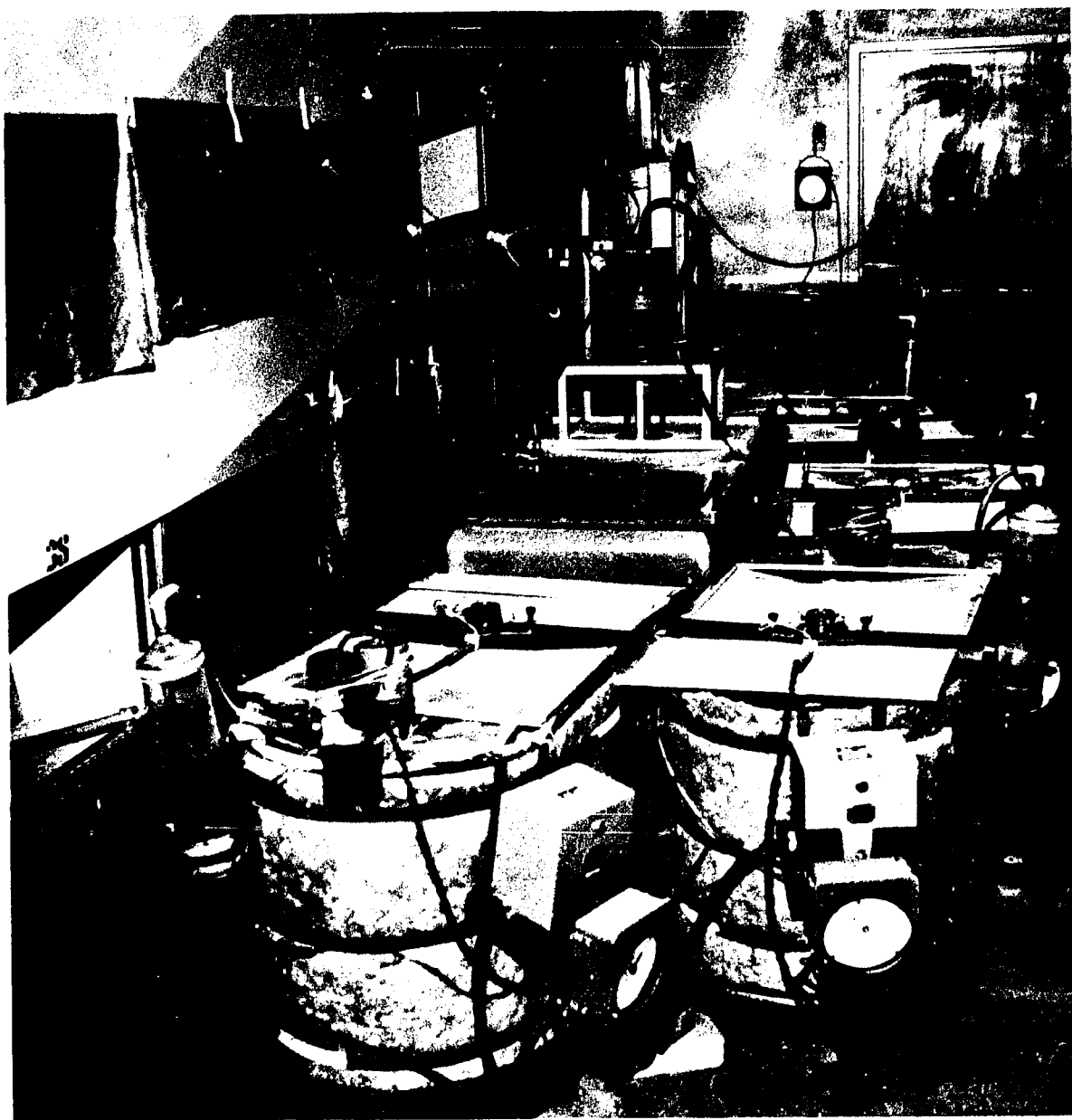


Figure 1. Utah State University Fisheries Laboratory with controlled-temperature fish acclimation tanks.

were held at the test temperatures for at least 2 weeks before experiments began. Fish were fed twice daily in the laboratory on the same diet fed at the hatchery.

Water quality analyses were made on water from the Min-O-Cool reservoir and on water taken from each acclimation tank. Fish were maintained at the desired acclimation temperature for at least a week before water samples were collected. All analyses except temperature and dissolved oxygen were made by the Utah Water Research Laboratory (Table 1).

With the exception of the experiments conducted under controlled light and dark conditions, fish in the laboratory were exposed to continuous fluorescent light of 2 to 12 ft-c (depending on the location of the tanks) during acclimation and experimentation.

Apparatus

Apparatus used to determine active and standard metabolic rates will be referred to as the "active respirometer" and "standard respirometer," respectively. These were submersed in controlled-temperature water baths.

Water baths

Respirometers were located in two 400-liter constant-temperature water baths. Water was circulated between the baths and a large Min-O-Cool reservoir. Water levels throughout the system were controlled by pumps and float valves. Water temperatures in this system were controlled by buffering thermoregulator-controlled heaters against the

Table 1. Water analyses, Utah State University Fisheries Laboratory, Logan, Utah, 1967--expressed as parts per million unless stated otherwise

	Temperature (C)							
	5	5	10	15	20	25	8-10 ^a	8-10 ^a
	Date							
	Oct. 17	Oct. 25	Oct. 11	Sept. 14	Oct. 11	Oct. 18	Oct. 25	Sept. 14
Color (units)	<5	<5	<5	<5	<5	<5	<5	<5
Turbidity (JTU)	<25	<25	25	<25	<25	<25	<25	<25
Dissolved oxygen	10.0	10.0	8.9	7.8	7.1	6.7	9.3	9.3
Total hardness as CaCO ₃	185	190	190	190	210	185	185	160
Alkalinity total	217.1	218.3	214.7	218.3	212.2	218.3	217.1	213.5
Bicarbonate	217.1	218.3	214.7	218.3	212.2	218.3	217.1	213.5
Carbonate	0	0	0	0	0	0	0	0
Chloride as Cl	7.5	6.5	5.0	4.5	5.0	7.0	6.0	3.5
Calcium as Ca	7.7	32.0	12.2	23.4	9.6	5.3	35.2	18.2
Magnesium as Mg	40.3	26.8	38.8	32.0	40.3	41.8	23.6	27.8
Sodium as Na	1.2	1.8	1.2	1.2	1.2	1.2	1.8	1.2
Potassium as K	0.3	0.3	0.4	0.6	0.4	0.3	0.3	0.6
Sulphate as SO ₄	7.75	9.60	8.50	7.25	8.75	8.75	9.60	6.70
Ammonia as NH ₃	0.55	0.46	0.72	0.28	0.57	0.78	0	0.12
Phosphate as PO ₄	0.212	0.197	0.340	0.312	0.190	0.235	0.020	0.005
Nitrate as NO ₃	0	0.35	0.20	0.65	0.72	0.15	0.10	1.00
Silica as SiO ₂	2.9	3.8	3.4	2.9	3.5	3.1	3.8	3.6
Electrical conductivity (µmhos/cm)	320	325	290	318	290	320	315	310
Total dissolved solids	167	177	180	207	199	173	179	158
pH	8.7	8.3	8.2	7.4	8.1	8.8	8.2	7.5

^aMin-O-Cool reservoir

Min-O-Cool refrigeration unit. A recording thermometer continuously monitored the water temperature. Water temperatures in the active respirometer bath were controlled to within ± 0.3 C, while temperatures in the standard respirometer bath seldom fluctuated by more than 0.1 C.

Active respirometer

The prototype of this apparatus was described by Blazka, Volf, and Cepela (1960). Smit (1965) built an apparatus similar to the Blazka respirometer. The respirometer used in the present study was slightly modified from that reported by Smit (1965; pers. comm.). It consisted of two clear acrylic plastic chambers in which water was drawn through the inner cylinder by a propeller, and returned between the outer and inner cylinders (Figures 2 and 3). The outer cylinder was 6.4 mm thick, 76.2 cm long, and 22.2 cm in diameter. The inner cylinder measured 3.2 mm thick, 6.1 cm long, with a diameter of 15.9 cm. A stainless steel screen closed off one end of the inner cylinder. The plastic ends of the respirometer were 15.9 mm thick and were fitted with large O-ring seals to prevent leakage. The ends were held by six wing-nuts and could be easily removed for cleaning and maintenance of the respirometer. A rectangular opening 10.2 cm wide and 14.0 cm long in the top of the two cylinders permitted insertion and removal of fish. A lid, custom-made to fit the openings in the cylinders, restored the convex shape of the cylinder walls and sealed the respirometer. Vertical plastic inlet and outlet tubes 15.2 cm high and 5.1 cm in diameter extended from the top of the

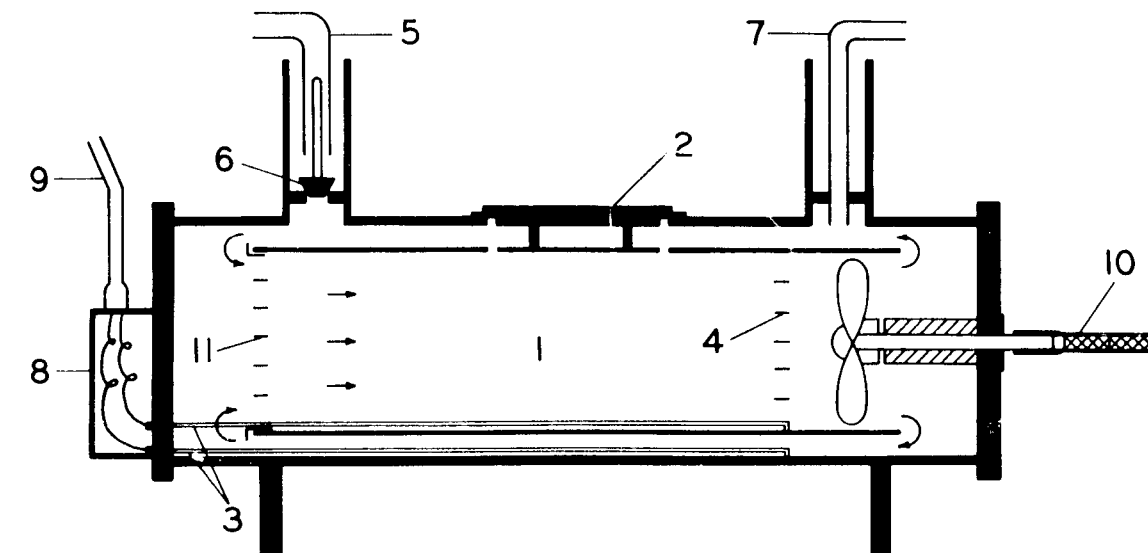


Figure 2. Diagrammatic sketch of the active respirometer.

- 1. Fish chamber. 2. Lid. 3. Electrodes. 4. Metal grid.
- 5. Inlet hose. 6. Inlet stopper. 7. Outlet hose.
- 8. Waterproof box for electrode power connection. 9. Power supply to electrodes. 10. Flexible drive shaft. 11. Fish chamber screen.

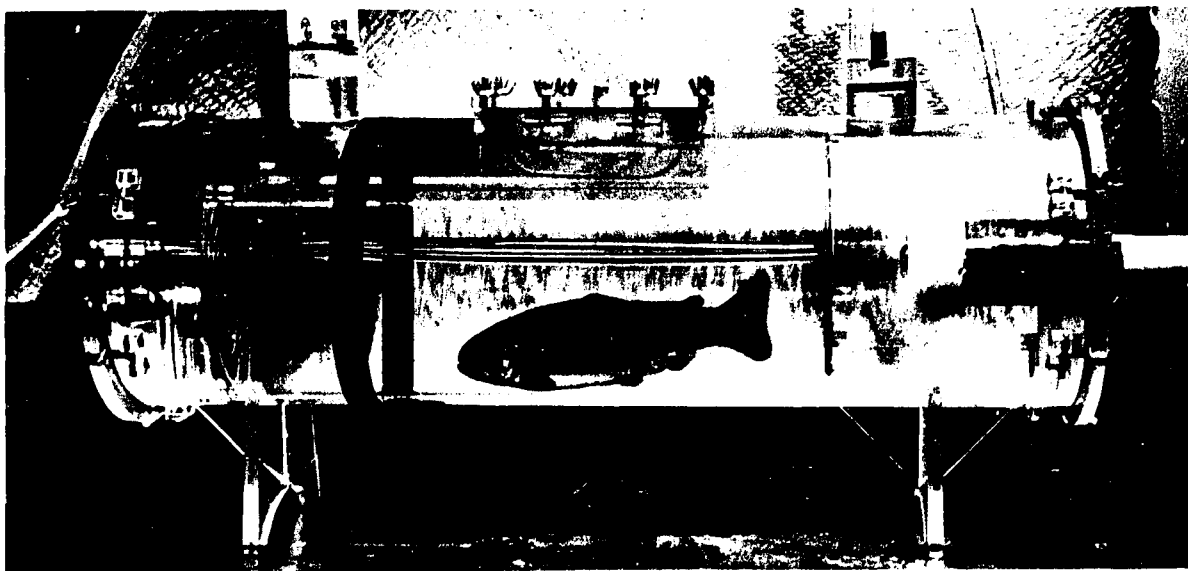


Figure 3. Rainbow trout swimming in the active respirometer (Apparatus removed from water bath for photographic purposes).

respirometer at either end to permit flushing of the chamber. A black band of plastic tape was wrapped around the respirometer to provide a reference site for the swimming fish.

The propeller and propeller shaft were supported by a neoprene bearing and seal at one end of the respirometer, and extended a short distance into the inner cylinder. The propeller shaft was connected to a flexible drive driven by a one-eighth H.P. variable-speed electric motor. A relatively uniform flat velocity profile was produced over the range of velocities at which fish were forced to swim (61-107 cm/sec). This was achieved by operating the propeller in reverse and pulling the water through the inner cylinder. Sawdust chips placed in the flow were used to observe the flow pattern. Velocity measurements to the nearest 3.0 cm/sec were made at 2.5 cm intervals across the diameter of the inner cylinder with a pitot tube inserted through the respirometer lid. Velocity profiles were determined for various units on the propeller-motor speed control. The velocity for any given unit on the speed control was expressed as the mean of all measurements for that particular velocity profile (Table 2).

Two stainless-steel electrodes were attached to the opposite end of the respirometer. Power supply leads to the electrodes were enclosed in a waterproof box fitted to the end of the respirometer. The electrodes extended into the respirometer, one on either side of the respirometer wall, to a point opposite a stainless-steel grid located 4 cm from the propeller. The electrodes were connected to a variable transformer and provided the

Table 2. The relation of propeller-motor speed control units to mean velocity profiles, active respirometer

Motor speed control units	Mean water velocity (cm/sec)
100	107
95	104
90	91
85	91
80	76
75	73
70	67
65	58
60	43
55	40

grid with an applied charge of approximately 8-10 v a-c.

The respirometer rested on a plastic stand in the water bath. Only the uppermost parts of the inlet and outlet tubes extended out of the water. Submersion of the apparatus prevented water bubbles from accumulating in the chamber.

Fish were forced to swim against the current in that portion of the inner cylinder in front of the electric grid referred to as the fish chamber (Figure 3). The swimming speed of the fish was dependent upon the speed of the propeller. The stimulus provided by the electric grid caused avoidance without strong reaction, and fish quickly learned to occupy positions in the fish chamber upstream from the electric grid with their heads at the black orientation band.

Standard respirometer

Standard metabolism measurements were made with apparatus resembling that described by Beamish and Mookherjee (1964). It consisted of a respiration chamber fitted with a glass flowmeter (Figure 4). The flowmeter operated on a heat-loss principle and was used to detect and estimate the activity of fish within the chamber.

The respiration chamber consisted of a modified thin-walled, stainless-steel, sauce pot. The chamber was 21 cm deep with a diameter of 30 cm. The volume of the chamber was 12.6 liters. Inlet and outlet tubes were fitted to the chamber for flushing and obtaining water samples, and the cover was permanently attached and modified to accommodate the flowmeter. A 15.5 x 10.0 cm hole was cut in the top of the respirometer for insertion and removal of fish. A metal collar around the opening was fitted with a plastic lid and rubber seal. A 5.5 cm diameter metal tube 16.2 cm long extended upward from the top of the respirometer and enclosed the flowmeter.

A flowmeter described by Beamish and Mookherjee (1964) was obtained from Dr. F. E. J. Fry, Department of Zoology, University of Toronto, Canada, and others were made from this design (Figure 5). The flowmeter tube extended down into the chamber to within 13 mm of the bottom and was supported by a close-fitting collar. The flowmeter was a thermoregulator with a small electric heater of approximately 5 w inserted into a blind tube in a mercury bulb. A relay in the heater circuit was controlled by the mercury thermoregulator (Figure 6). The heater maintained

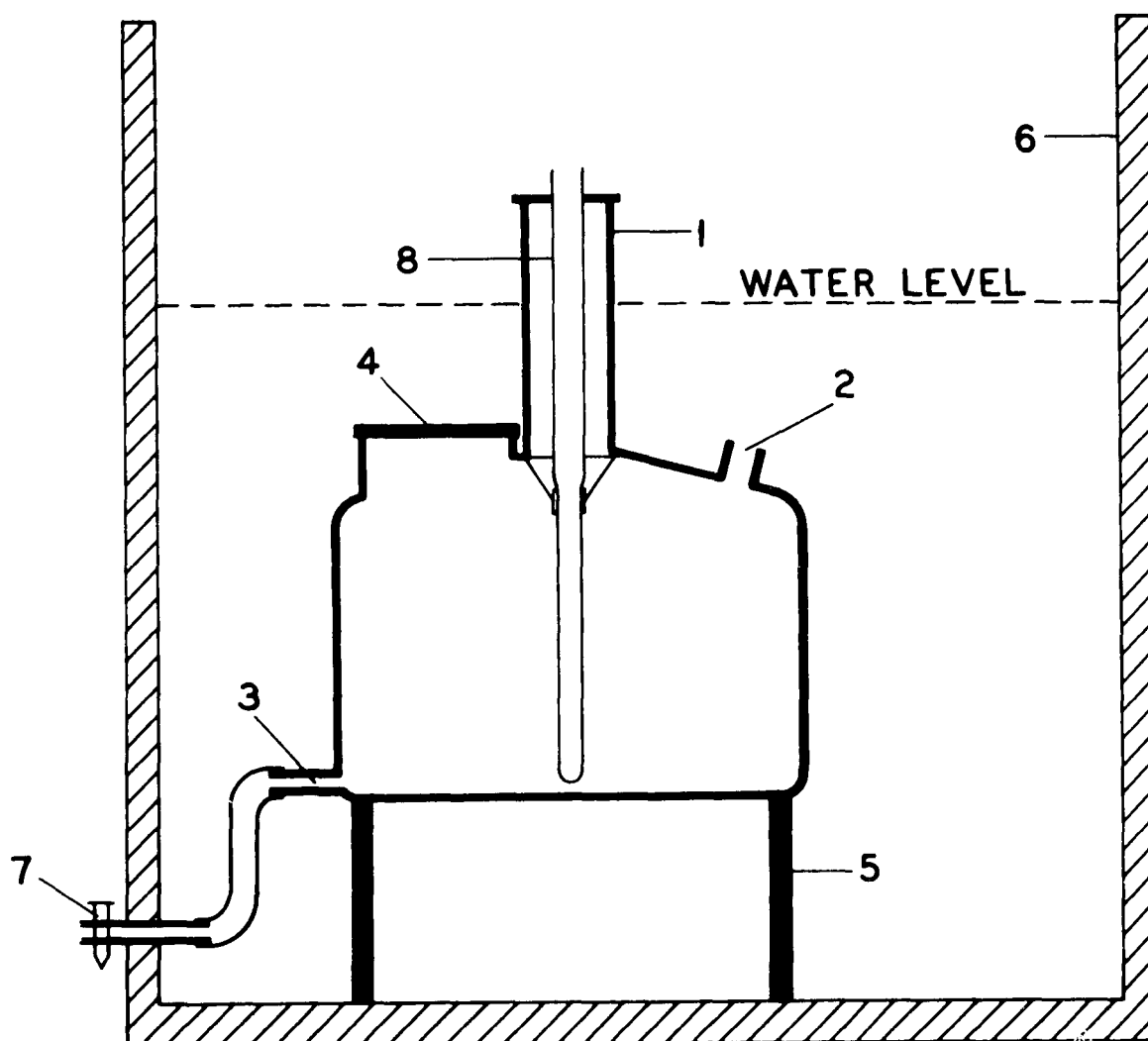


Figure 4. Diagrammatic sketch of the standard respirometer. 1. Flow-meter tube. 2. Inlet tube. 3. Outlet tube. 4. Plastic lid for insertion and removal of fish. 5. Stainless-steel support for respirometer. 6. Insulated side of water bath. 7. Spring clamp. 8. Flowmeter.

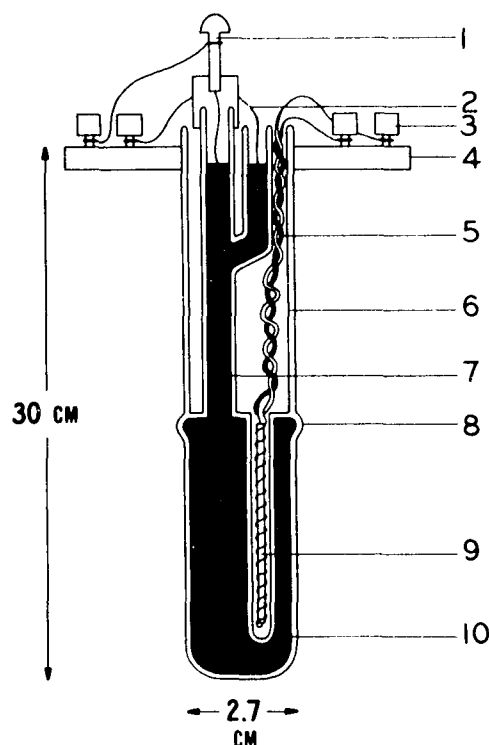


Figure 5. Diagrammatic sketch of the flowmeter. 1. Adjustable electrode for baseline setting. 2. Immersed electrode. 3. Contact terminal. 4. Lucite platform. 5. Insulated electrical wire. 6. Glass wall. 7. Mercury-filled capillary tube. 8. Projection for support of flowmeter in the respiration chamber. 9. Heater inserted in blind tube. 10. Mercury-filled bulb. (After Beamish and Mookherjee, 1964)

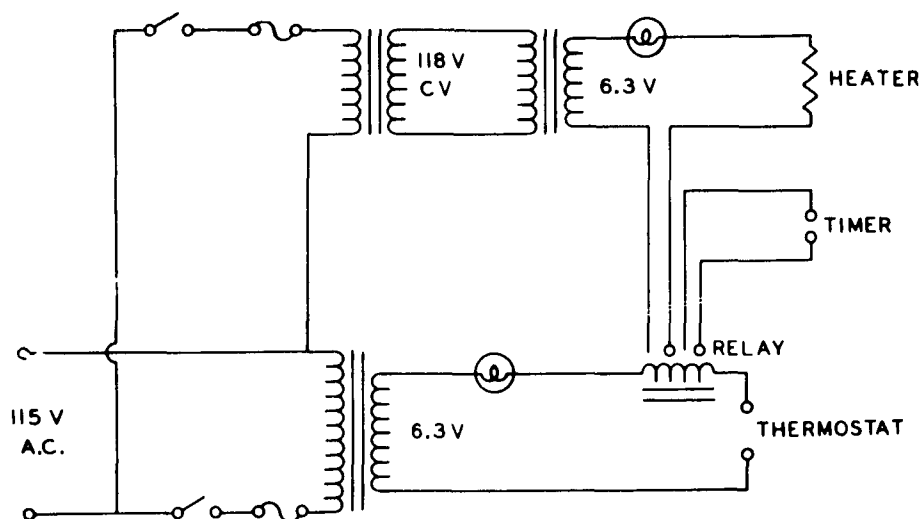


Figure 6. Circuit diagram, flowmeter control.

the mercury bulb at a constant temperature, slightly above that of the water in the respiration chamber. The contacts were adjusted so that the heater circuit was closed approximately 50 per cent of the experimental period. A constant-voltage transformer prevented line-voltage fluctuations from changing the heater's wattage and affecting the operation of the flowmeter. The length of time the heater was in circuit was measured in seconds by a manual reset electrical timer.

Swimming of a fish in the chamber induced turbulence which increased heat loss from the flowmeter bulb. This increased the time that the heater was in circuit, and it was this increase in time that was equated to the fish's activity. The time that the heater was in circuit in the absence of a fish was expressed as percentage of the experimental period and was called the "baseline." When a fish was in the chamber, activity was expressed as the percentage time increase in heater operation above the baseline setting.

The respirometer was mounted on three legs and submerged in the water bath to a point midway up the flowmeter tube. Air-saturated water entered the inlet tube of the respirometer from the water bath. The outlet tube extended beyond the water bath by a hose, which was pinched off with a spring clamp. Water samples were obtained from the outlet hose. The respirometer was flushed by a small electric pump that forced water into and out of the chamber through the inlet tube at a rate of approximately 1.5 liters per minute. Fish appeared indifferent to the insert, presence, or removal of the pump-flushing tube from the darkened chamber.

Light control

To provide light and dark periods to coincide with the natural photoperiod, a light-proof enclosure was erected around two acclimation tanks, and the photoperiod in the enclosure was controlled automatically by time switches (Figure 7). The light sources consisted of two 40 w incandescent bulbs, one suspended at each end of the enclosure, and four fluorescent bulbs 2.4 m long extending almost the entire length of the enclosure. Light intensities of from 28 to 60 ft-c at the surface of the water in the acclimation tanks were provided by the 40 w bulbs. This light and light from the fluorescent bulbs provided 380 to 450 ft-c light intensity at the surface of the water in the tanks. Light intensities varied because of the relative position of the lights over the tanks. Fish were exposed to periods of different light intensities as follows: 30 minutes of low intensity illumination (28-60 ft-c)--followed by 9 hours of high intensity lighting (380-450 ft-c)--followed by a second 30-minute exposure period of low intensity lighting only. Fish were then in darkness for 14 hours.

Experimental Procedures

General

A different fish was used for each measurement of active and standard metabolism. Fish were starved for 24 hours before transfer to the respirometers to avoid undue deposition of feces in the chambers. Unless otherwise stated, all measurements of metabolism were made between 7:00 AM and 2:00 PM.

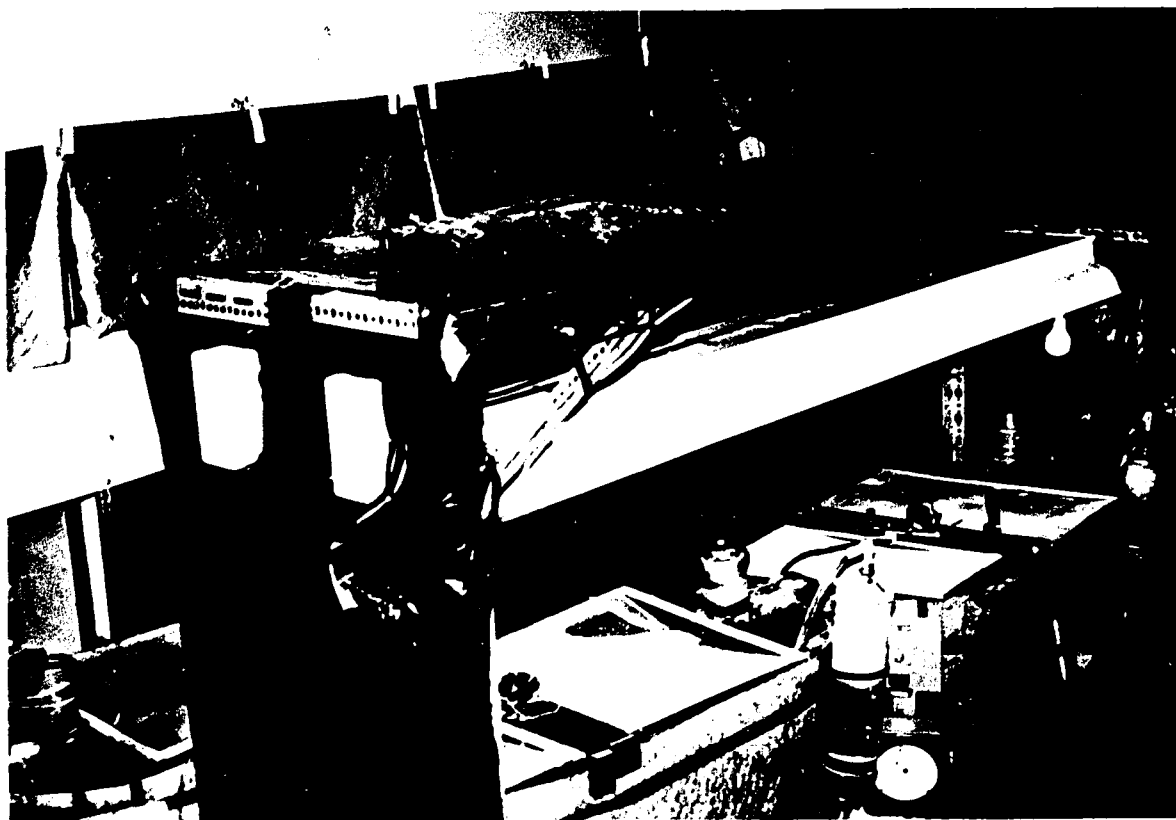


Figure 7. Apparatus used for controlled light-and-dark experiments.

One 300-ml water sample was collected from the respirometers immediately before and after each test period. The difference in dissolved-oxygen concentration between the samples was the amount of oxygen consumed by the fish. Reduction in oxygen caused by the fish during the test period was usually about 1.0 ppm and seldom exceeded 2.0 ppm. In order to maintain dissolved-oxygen concentrations at relatively high levels at all times, the length of active and standard metabolism experiments was varied according to the water temperature. Dissolved-oxygen concentration was seldom less than 6 ppm and rarely dropped to 5 ppm. Following each experiment fish were dried with absorbent paper, weighed, measured, and sexed.

Active metabolism

Water was circulated in the active respirometer at very low velocity immediately before each experiment to orient each new fish. Fish were allowed a 15-20 minute adjustment period while the chamber was being flushed to allow dissolved-oxygen concentrations to reach air saturation. A water sample was collected from the outlet hose, the outlet hose was closed, and the inlet tube stoppered. The speed of the propeller was increased slowly in 1-2 minutes until water velocities reached 61-107 cm/sec depending on the test temperature. Active metabolism experiments at 5 and 10 C, and those at 15, 20, and 25 C lasted 20 and 15 minutes, respectively. At the end of the forced swimming period, the propeller speed was reduced to a very slow speed, the stopper removed from the

inlet tube, and a water sample collected from the outlet hose. The fish was removed from the chamber. Fish unable to swim at the test velocity for the required time were discarded.

Standard metabolism

A baseline reading was obtained immediately before insertion of the fish into the respirometer. A black plastic cover was fitted over the transparent lid of the chamber and precautions were taken to suppress external stimuli that might excite the fish or stimulate it to activity (Fry, 1957). Fresh water was continually pumped into the chamber. Approximately 12-14 hours elapsed before the experiment started. A water sample was collected, the outlet hose closed, the flushing tube removed from the chamber, and the inlet tube stoppered. Individual test periods to determine oxygen consumption and activity lasted 40 minutes at 5 C, 30 minutes at 10, 15, and 20 C, and 15 minutes at 25 C. At the end of each test period, the stopper was removed from the inlet tube and a water sample was collected from the outlet hose; the time that the heater was in operation was recorded; the flushing tube was reinserted; and water was pumped into the chamber for 20 minutes to restore the original concentration of dissolved oxygen (usually reached within 10 to 15 minutes).

Depending on the temperature, an experiment lasted 5 to 7 hours to allow six determinations of oxygen consumption and activity. Following removal of the fish from the chamber, an end baseline reading was obtained. Baseline determinations made before and after each experiment rarely

differed by more than 60 seconds (6.7 per cent of the shortest test period). Results were discarded when greater differences occurred. The six determinations of oxygen consumption and activity were plotted on a semi-logarithmic grid. A straight-line relationship was obtained by plotting the logarithm of oxygen consumption against activity. Standard metabolism was estimated by extrapolation of this line to zero activity.

Starvation and controlled light and dark

Procedures employed in studies of the effects of starvation and time of day on respiratory metabolism were essentially the same as those described above. Starvation experiments involved depriving fish of food for 1, 2, 3, 6, and 9 days. The 24-hour starvation period before the experiments began was considered as the first day of starvation. Active metabolism measurements were made from 7:00 AM to 5:00 PM and measurements of standard metabolism were taken from 7:00 AM to 2:00 PM.

The effect of time of day on active and standard metabolism was investigated during morning (7:00 AM-1:00 PM) and afternoon (1:00 PM-7:00 PM). Active and standard metabolic rates were determined for fish subjected to controlled light and dark periods, and continuous laboratory illumination.

Dissolved oxygen determinations and water sampling techniques

The Alsterberg azide modification of the Winkler method (American Public Health Association, 1960) was used to determine dissolved-oxygen

concentration. This modification was employed as a precautionary measure against nitrite interference in oxygen determinations. Two oxygen determinations were made immediately on each water sample. If results differed by more than 0.1 ml sodium thiosulphate (0.01 N), a third titration was done and the three measurements averaged. However, the first two titrations rarely differed by more than 0.1 ml sodium thiosulphate.

Water was flushed repeatedly through the sample bottle before a water sample was collected at the beginning of each test. At the end of the test period approximately 500 ml of water was removed from the respirometers to flush the sample bottle and obtain the water sample. Time required to collect this water sample was between 10 and 15 seconds. As fresh water was being added to the chamber while the water sample was collected, it was necessary to determine if the dissolved oxygen concentration was affected by the addition of fresh water. Dissolved oxygen in three 300-ml water samples collected consecutively from the respirometers did not differ by 0.1 ppm. As less than this volume of water was required for each sample, the differences in all cases were considered negligible.

Data Analyses and Treatment of Results

Four hundred and twenty-two active and 196 standard metabolism measurements were made from October, 1966 to December, 1967. Because of numerous preliminary experiments, measurements of active metabolism exceeded those of standard metabolism.

Semi-logarithmic computer regression analyses of $\log_{10}$ oxygen consumption and activity measurements gave estimates of standard metabolism.

Covariance analyses adjusted weights of fish to a common mean, and, unless otherwise indicated, these and other statistical results are reported at the 5 per cent level of significance (Snedecor, 1956). In some instances, statistical analyses without covariance were employed to compare scope for activity values, and standard and active metabolic rates. However, these respiratory rates were compared only when mean weights of fish were not significantly different at the 5 per cent level. Based on the $\underline{z}$ distribution reported by Ostle (1963), respiratory rates were significantly different at the 5 per cent level if the calculated $\underline{z}$ value exceeded 1.96.

Rates of respiratory metabolism have been reported in the text as mean rates plus or minus one standard error of the mean.

RESULTS AND OBSERVATIONS

Preliminary Tests

Effect of electricity and acrylic plastic on dissolved oxygen

Fry (1958) indicated that dissolved-oxygen concentrations might be affected by an alternating electrical current. Beamish (1967, pers. comm.) reported that some acrylic plastic materials absorb oxygen from water. In September, 1966 preliminary experiments with the active respirometer were conducted with and without power supplied to the electric grid. Experiments lasted 15 minutes and water was circulated in the apparatus at 91 cm/sec. Dissolved-oxygen concentrations in water samples collected from the respirometer immediately before and after the experiments differed by less than 0.1 ppm, which indicated that effects of electricity and plastic on dissolved oxygen concentration were negligible.

Effect of fish-holding facilities on respiratory metabolism

The effect of confinement in 400-liter acclimation tanks upon active metabolism of Hull-Erickson trout was studied from October 26 to November 5, 1966. Active metabolic rates were determined for 21 fish maintained in a 400-liter acclimation tank at the Utah State Experimental Fish Hatchery and were compared to similar measurements of 23 fish held in a hatchery raceway (Table 3). An electric aerator pump circulated water

Table 3. Active metabolism of Hull-Erickson trout acclimated to 17 C and maintained in an acclimation tank or hatchery raceway

Holding facilities		Fork length (cm)	Wet weight (g)	Sex		Active metabolism (mg/kg hr)	Swimming speed	
				M	F		(cm/sec)	(L/sec) ^a
Acclimation tank	Number	(21)	(21)	7	14	(21)	(21)	(21)
	Range	20.2-28.0	97.3-273.9	-	-	468-738	91	3.3-4.5
	Mean	24.0	169.1	-	-	559	-	3.9
	S.D.	1.8	38.5	-	-	72	-	0.5
	S.E.	0.4	8.4	-	-	16	-	0.1
Hatchery raceway	Number	(23)	(23)	8	15	(23)	(23)	(23)
	Range	23.8-30.5	167.9-341.6	-	-	384-630	91	3.0-3.8
	Mean	27.0	249.9	-	-	468	-	3.4
	S.D.	1.9	51.5	-	-	58	-	0.2
	S.E.	0.4	10.7	-	-	12	-	*

* < 0.05

^a Fish body lengths per sec.

in the tank, and flow through the tank was regulated to approximate rate of exchange in laboratory acclimation tanks. Temperature of the hatchery water was 17 C, and fish were exposed to the natural photoperiod. Fish were held in the acclimation tank for at least 6 weeks. No mortality occurred during this time. Fish from the acclimation tank and raceway were delivered to the laboratory and exercised at a velocity of 91 cm/sec (3.0-4.5 L/sec--fish body lengths per sec) in the active respirometer within 10 days.

The mean weights of fish held in the acclimation tank and hatchery raceway were 169.1 and 249.9 g, respectively, and were significantly different. As a production diet was not determined for fish in the acclimation tank, presumably these fish were underfed, and were smaller than those in the raceway. Mean active metabolic rates of fish held in the acclimation tank and hatchery raceway were 559 ± 16 and 468 ± 12 mg/kg hr, respectively, and were significantly different. However, when fish weights were corrected to a common mean by analyses of covariance, mean active metabolic rates of the two groups were not significantly different ($F=3.95$ with 1 and 41 d.f.).

Effect of adjustment time in the respirometer upon active metabolism

During October 11 to November 11, 1966, a study was made to determine if active metabolic rates were affected by conditioning fish to the active respirometer for various periods of time before experiments. Hull-Erickson trout were conditioned to the respirometer by exercising them at

approximately 15 cm/sec velocity for 15 minutes, and 2 to 5 hour periods. Mean weights of fish conditioned for 15 minutes, and 2 to 5 hour periods were 214.8 and 230.6 g, respectively. Active rates of oxygen consumption were determined for fish acclimated to 10 C and forced to swim at 91 cm/sec (3.1-4.4 L/sec) (Table 4).

Mean active metabolic rates of 16 fish conditioned for 2 to 5 hours, and 21 fish conditioned for 15 minutes were 426 ± 17 and 450 ± 17 mg/kg hr, respectively, and were not significantly different ($F=0.59$ with 1 and 34 d.f.).

Effect of increased temperature and water velocity on active metabolism

The effect of increased water velocity and temperature on active metabolism was studied on 70 small (60.1-146.6 g) and 55 large (134.6-241.4 g) Hull-Erickson trout. Experiments were conducted from February 1 to April 4, 1967. Fish were acclimated to 5, 10, 15, 20, and 25 C and exercised at 43 to 107 cm/sec velocities (1.5-5.1 L/sec). Where possible, at least two measurements of active metabolism were made for small and large fish at increased velocities for each of the five acclimation temperatures (Appendix). Results indicated that fish may respire at active metabolic rates when forced to swim at speeds as low as 1.5-2.4 L/sec. This is not surprising because at low velocity, excitement induced by handling immediately before experiments (Keys, 1930; Wells, 1932; Black, Fry, and Scott, 1939), and stimulation by an electric field (Basu, 1959) can induce active rate of oxygen consumption.

Table 4. Active metabolism of Hull-Erickson trout acclimated to 10 C and pre-conditioned to the respirometer

Conditioning period		Fork length (cm)	Wet weight (g)	Sex		Active metabolism (mg/kg hr)	Swimming speed	
				M	F		(cm/sec)	(L/sec) ^a
15 min.	Number	(21)	(21)	15	6	(21)	(21)	(21)
	Range	21.0-29.7	116.0-353.8	-	-	300-606	91	3.1-4.4
	Mean	25.3	214.8	-	-	450	-	3.8
	S.D.	2.4	66.1	-	-	76	-	0.4
	S.E.	0.5	14.4	-	-	17	-	0.1
2-5 hr.	Number	(16)	(16)	8	8	(16)	(16)	(16)
	Range	21.0-29.7	105.6-320.2	-	-	300-510	91	3.1-4.4
	Mean	25.9	230.6	-	-	426	-	3.6
	S.D.	2.5	67.4	-	-	66	-	0.4
	S.E.	0.6	16.9	-	-	17	-	0.1

^aFish body lengths per sec.

However, Job (1955) stated that the mere handling of fish is not always sufficient to induce maximum oxygen consumption, but that fish must be forced to swim and exert themselves to obtain this value.

Temperature and Respiratory Metabolism of Rainbow Trout

The effect of temperature on standard and active metabolism of Hull-Erickson and De Smet trout was studied at 5, 10, 15, 20, and 25 C. Fish acclimated to 25 C refused to feed at this high temperature and were generally smaller than others.

Active metabolism

Hull-Erickson strain. From April 22 to June 24, 1967, 54 measurements of active metabolism were made on fish that weighed 136.7 to 248.6 g and were 23.0 to 28.5 cm long (Table 5). Swimming speeds varied from 2.3 to 3.8 L/sec depending on the water temperature and size of fish. Fish at 10, 15, and 20 C were exercised at a velocity of 91 cm/sec. At 5 and 25 C, the majority of fish were unable to swim at 91 cm/sec velocity and were exercised at lower velocities. Active metabolic rates increased from a mean of 384 ± 15 mg/kg hr at 5 C to 468 ± 11 mg/kg hr at 10 C, and reached a maximum of 576 ± 22 mg/kg hr at 15 C. Active metabolism was only slightly decreased at 20 C but was markedly reduced to 474 ± 18 mg/kg hr at 25 C.

Temperature had a significant effect on active metabolic rates at the 1 per cent level ($F=20.05$ with 4 and 48 d.f.). Active rates of oxygen

Table 5. Active metabolism of Hull-Erickson trout acclimated to different temperatures

Acclimation temperature (C)		Fork length (cm)	Wet weight (g)	Sex		Active metabolism (mg/kg hr)	Swimming speed	
				M	F		(cm/sec)	(L/sec) ^a
5	Number	(10)	(10)	3	7	(10)	(10)	(10)
	Range	24.5-28.5	190.2-248.6	-	-	318-474	67	2.3-2.7
	Mean	26.8	219.4	-	-	384	-	2.5
	S.D.	1.4	20.8	-	-	46	-	0.5
	S.E.	0.4	6.6	-	-	15	-	0.2
10	Number	(10)	(10)	3	7	(10)	(10)	(10)
	Range	25.3-27.5	175.1-242.1	-	-	420-540	91	3.3-3.6
	Mean	26.8	214.7	-	-	468	-	3.5
	S.D.	0.8	21.2	-	-	36	-	*
	S.E.	0.3	6.7	-	-	11	-	*
15	Number	(10)	(10)	4	6	(10)	(10)	(10)
	Range	25.2-27.0	187.0-240.9	-	-	468-672	91	3.4-3.6
	Mean	26.0	209.6	-	-	576	-	3.5
	S.D.	0.6	16.5	-	-	68	-	*
	S.E.	0.2	5.2	-	-	22	-	*

Table 5. Continued

Acclimation temperature (C)		Fork length (cm)	Wet weight (g)	Sex		Active metabolism (mg/kg hr)	Swimming speed	
				M	F		(cm/sec)	(L/sec) ^a
20	Number	(10)	(10)	1	9	(10)	(10)	(10)
	Range	23.7-27.5	149.8-239.8	-	-	456-732	91	3.3-3.8
	Mean	25.3	193.1	-	-	570	-	3.6
	S.D.	1.4	32.7	-	-	68	-	0.2
	S.E.	0.4	10.3	-	-	22	-	0.1
25	Number	(14)	(14)	8	6	(14)	(14)	(14)
	Range	23.0-25.8	136.7-192.6	-	-	384-606	73-91	2.8-3.8
	Mean	24.7	161.1	-	-	474	79	3.3
	S.D.	0.9	16.1	-	-	67	15	0.3
	S.E.	0.2	4.3	-	-	18	3	0.1

* <0.05

^aFish body lengths per sec.

consumption at 5 C differed significantly from those at other temperatures (Table 6). Active rates of oxygen uptake at 10 C differed significantly from those at other temperatures except 25 C.

Table 6. Active metabolic rates of Hull-Erickson trout at different temperatures--expressed as mg/kg hr (Values joined by underscore are not significantly different from each other at the 5 per cent level, Newman-Keul's multiple range test)

	Temperature				
	5	10	25	20	15
No. of observations	(10)	(10)	(14)	(10)	(10)
Mean active rates	384	<u>468</u>	<u>474</u>	<u>570</u>	<u>576</u>

De Smet strain. Sixty-five measurements of wild-trout active metabolism were made from September 28 to October 30, 1967. Fish weighed 78.5 to 397.8 g and were 19.8 to 30.0 cm long (Table 7). Fish swam at 76 to 104 cm/sec velocities (2.9-4.7 L/sec) with relatively little difficulty at each of the test temperatures. Fish acclimated to 5 C swam at slightly lower velocities than fish acclimated to higher temperatures. Active metabolism increased with increasing temperature from a minimum of 366 ± 16 mg/kg hr at 5 C to a maximum of 592 ± 19 mg/kg hr at 20 C. Active metabolism was decreased only slightly at 25 C to 588 ± 19 mg/kg hr.

Active metabolism of De Smet trout was significantly affected by temperature at the 1 per cent level ($F=23.70$ with 4 and 59 d.f.). The

Table 7. Active metabolism of De Smet trout acclimated to different temperatures

Acclimation temperature		Fork length (cm)	Wet weight (g)	Sex		Active metabolism (mg/kg hr)	Swimming speed	
				M	F		(cm/sec)	(L/sec) ^a
5	Number	(15)	(15)	6	9	(15)	(15)	(15)
	Range	22.5-30.0	135.7-299.5	-	-	270-522	76-104	2.9-4.0
	Mean	25.8	200.2	-	-	366	91	3.5
	S.D.	2.2	53.3	-	-	62	9	0.6
	S.E.	0.6	13.8	-	-	16	3	0.2
10	Number	(15)	(15)	7	8	(15)	(15)	(15)
	Range	22.0-29.5	99.1-397.8	-	-	390-636	91-104	3.2-4.7
	Mean	25.7	204.9	-	-	492	98	4.0
	S.D.	2.4	74.1	-	-	78	6	0.4
	S.E.	0.6	19.1	-	-	20	2	0.1
15	Number	(10)	(10)	7	3	(10)	(10)	(10)
	Range	22.2-29.2	105.7-275.5	-	-	348-720	91	3.1-4.2
	Mean	24.7	177.7	-	-	552	-	3.6
	S.D.	2.3	57.3	-	-	118	-	0.4
	S.E.	0.7	18.1	-	-	37	-	0.1

Table 7. Continued

Acclimation temperature		Fork length (cm)	Wet weight (g)	Sex		Active metabolism (mg/kg hr)	Swimming speed	
				M	F		(cm/sec)	(L/sec) ^a
20	Number	(14)	(14)	9	5	(14)	(14)	(14)
	Range	21.5-29.0	90.5-285.0	-	-	486-756	91-104	3.1-4.3
	Mean	25.1	177.8	-	-	592	98	3.6
	S.D.	2.1	47.8	-	-	70	6	0.4
	S.E.	0.6	12.8	-	-	19	2	0.1
25	Number	(11)	(11)	4	7	(11)	(11)	(11)
	Range	19.8-30.0	78.5-291.9	-	-	444-666	91	3.1-4.6
	Mean	24.4	155.6	-	-	588	-	3.9
	S.D.	2.7	57.6	-	-	64	-	0.4
	S.E.	0.8	17.4	-	-	19	-	0.1

^aFish body lengths per sec.

mean active metabolic rate at 5 C differed significantly from mean rates at the other temperatures (Table 8). Mean active rates at 5 and 10 C were significantly different from those at the higher temperatures.

Table 8. Active metabolic rates of De Smet trout at different temperatures--expressed as mg/kg hr (Values joined by underscore are not significantly different from each other at the 5 per cent level, Newman-Keul's multiple range test)

	Temperature				
	5	10	15	20	25
No. of observations	(15)	(15)	(10)	(14)	(11)
Mean active rates	366	492	<u>552</u>	592	588

Hull-Erickson vs De Smet strain. Temperature effects were illustrated by plotting oxygen consumption rates on a semi-logarithmic grid (Figure 8). This tended to emphasize the relative rate of change in rate of oxygen consumption at different temperatures. Active metabolic rates of hatchery and wild trout increased with temperature from 5 to 15 C. Active metabolism of the two strains remained relatively unchanged with temperature increase to 20 C. Acclimation to 25 C resulted in a significant decrease in rate of active metabolism of the hatchery trout strain, while active metabolism of the wild trout remained unaffected. Mean rates of active metabolism for the two strains did not differ significantly

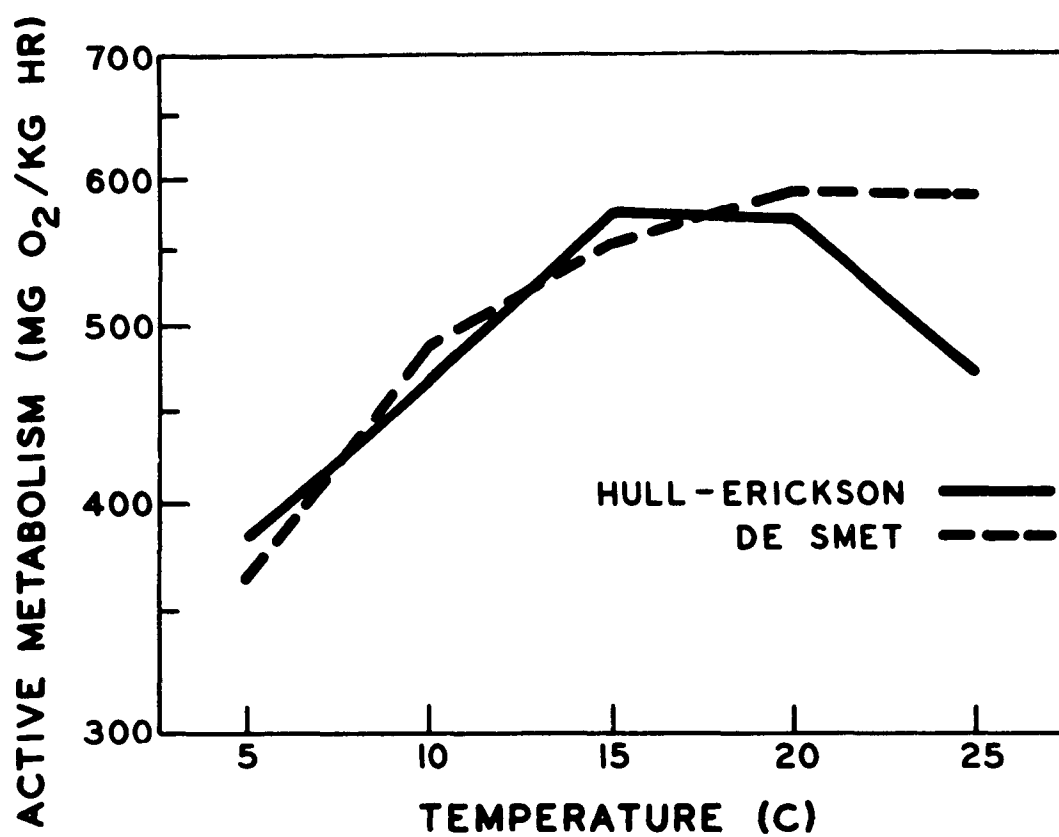


Figure 8. Temperature and active metabolism, rainbow trout, Hull-Erickson and De Smet.

at 5, 10, 15, and 20 C ($\underline{z} \geq 1.05$), but were significantly different at 25 C ($\underline{z}=4.35$).

Standard metabolism

Hull-Erickson strain. During April 17 to June 29, 1967, 48 standard metabolism measurements were made using fish that weighed 147.6 to 264.0 g and were 23.5 to 29.0 cm long (Table 9). Standard oxygen consumption increased with increasing temperature from a minimum of 36 ± 2 mg/kg hr at 5 C to a maximum of 138.0 ± 12 mg/kg hr at 25 C. Standard metabolism showed a progressive increase with increase in the intermediate temperatures.

Temperature had a significant effect on standard metabolism at the 1 per cent level ($F=17.65$ with 4 and 42 d.f.). Standard metabolic rates at 5 and 10 C were significantly lower than those at higher temperatures (Table 10). Standard oxygen consumption did not increase significantly from 5 to 10 C. Standard metabolism was significantly higher at 25 C than at other temperatures.

De Smet strain. Standard metabolism experiments were conducted from September 25 to October 27, 1967, using 55 fish that weighed 85.2 to 395.7 g and were 20.8 to 31.8 cm long (Table 11). Standard oxygen consumption increased with increasing temperature from a minimum of 42 ± 2 mg/kg hr at 5 C to a maximum of 120 ± 18 mg/kg hr at 25 C. Standard metabolism was 54 ± 4 mg/kg hr at 10 C, 60 ± 5 mg/kg hr at 15 C, and increased significantly to 94 ± 10 mg/kg hr at 20 C.

Table 9. Standard metabolism of Hull-Erickson trout acclimated to different temperatures

Temperature (C)		Fork length (cm)	Wet weight (g)	Sex		Standard metabolism (mg/kg hr)
				M	F	
5	Number	(13)	(13)	6	7	(13)
	Range	24.5-29.0	172.3-264.0	-	-	24-54
	Mean	26.9	213.9	-	-	36
	S.D.	1.5	25.9	-	-	9
	S.E.	0.4	7.2	-	-	2
10	Number	(8)	(8)	4	4	(8)
	Range	23.5-28.0	179.6-230.7	-	-	24-60
	Mean	26.1	206.7	-	-	42
	S.D.	1.3	16.6	-	-	14
	S.E.	0.5	5.9	-	-	5
15	Number	(9)	(9)	1	8	(9)
	Range	24.0-26.7	163.0-204.0	-	-	42-108
	Mean	25.4	190.9	-	-	78
	S.D.	1.1	13.1	-	-	17
	S.E.	0.4	4.4	-	-	6

Table 9. Continued

Temperature (C)	Fork length (cm)	Wet weight (g)	Sex		Standard metabolism (mg/kg hr)
			M	F	
20	Number	(10)	1	9	(10)
	Range	24.0-27.0	-	-	48-120
	Mean	26.0	-	-	84
	S.D.	0.8	-	-	21
	S.E.	0.3	-	-	7
25	Number	(8)	2	6	(8)
	Range	23.8-25.5	-	-	96-192
	Mean	24.7	-	-	138
	S.D.	0.5	-	-	33
	S.E.	0.2	-	-	12

Table 10. Standard metabolic rates of Hull-Erickson trout at different temperatures--expressed as mg/kg hr (Values joined by underscore are not significantly different from each other at the 5 per cent level, Newman-Keul's multiple range test)

	Temperature				
	5	10	15	20	25
No. of observations	(13)	(8)	(9)	(10)	(8)
Mean standard rates	<u>36</u>	<u>42</u>	<u>78</u>	<u>84</u>	138

Standard metabolism was significantly affected by temperature ($F=6.17$ with 4 and 49 d.f.). Standard metabolic rates at 5, 10, and 15 C were significantly lower than those at higher temperatures (Table 12).

Hull-Erickson vs De Smet strain. Standard metabolic rates of hatchery and wild trout strains at different temperatures were compared on a semi-logarithmic grid (Figure 9). The logarithm of oxygen consumption plotted against temperature shows this relation to be linear over the range of temperatures investigated. Standard metabolism of wild and hatchery trout increased with increasing temperature. With the exception of 15 C ($z=2.29$), standard metabolic rates of the two strains were not significantly different at each of the test temperatures ($z \leq 1.94$).

Scope for activity

Hull-Erickson strain. Scope for activity increased with increasing temperature to a maximum of 498 ± 37 mg/kg hr at 15 C (Figure 10). Scope increased from 348 ± 16 mg/kg hr at 5 C to 426 ± 21 mg/kg hr at 10 C.

Table 11. Standard metabolism of De Smet trout acclimated to different temperatures

Temperature (C)		Fork length (cm)	Wet weight (g)	Sex		Standard metabolism (mg/kg hr)
				M	F	
5	Number	(11)	(11)	2	9	(11)
	Range	24.0-30.0	163.0-328.6	-	-	30-48
	Mean	27.2	225.9	-	-	42
	S.D.	1.7	50.1	-	-	6
	S.E.	0.5	15.1	-	-	2
10	Number	(10)	(10)	5	5	(10)
	Range	22.8-31.0	135.0-395.7	-	-	42-87
	Mean	26.7	235.4	-	-	54
	S.D.	2.5	75.3	-	-	12
	S.E.	0.9	23.8	-	-	4
15	Number	(11)	(11)	4	7	(11)
	Range	22.5-31.8	132.5-375.9	-	-	24-84
	Mean	27.1	225.7	-	-	60
	S.D.	2.7	70.1	-	-	18
	S.E.	0.8	21.1	-	-	5

Table 11. Continued

Temperature (C)		Fork length (cm)	Wet weight (g)	Sex		Standard metabolism (mg/kg hr)
				M	F	
20	Number	(12)	(12)	4	8	(12)
	Range	21.5-27.0	114.8-228.7	-	-	54-156
	Mean	24.8	172.3	-	-	94
	S.D.	1.8	36.9	-	-	33
	S.E.	0.5	10.7	-	-	10
25	Number	(11)	(11)	4	7	(11)
	Range	20.8-26.5	85.2-188.0	-	-	48-222
	Mean	24.6	148.1	-	-	120
	S.D.	1.8	33.5	-	-	60
	S.E.	0.5	10.1	-	-	18

Table 12. Standard metabolic rates of De Smet trout at different temperatures--expressed as mg/kg hr (Values joined by under-score are not significantly different from each other at the 5 per cent level, Newman-Keul's multiple range test)

	Temperature				
	5	10	15	20	25
No. of observations	(11)	(10)	(11)	(12)	(11)
Mean standard rates	<u>42</u>	<u>54</u>	<u>60</u>	<u>94</u>	<u>120</u>

Scope was 486 ± 20 mg/kg hr at 20 C and was lowest at 25 C (336 ± 23 mg/kg hr).

Scope for activity at 5 C was significantly less than scope at 10, 15, or 20 C ($\underline{z} \geq 4.03$). Scope for activity was not significantly different at 15 and 20 C. Scope at 10 C was significantly less than that at 15 and 20 C ($\underline{z} \geq 2.30$). Scope for activity at 25 C was not included in statistical tests because the mean weight of fish at this temperature was significantly less than mean weights of fish at other temperatures. The greatest difference in mean weights of fish was 60.8 g. If the mean weight of fish at 25 C were increased by this amount, scope for activity at this temperature would probably have been lower. Job (1955) studied the effect of weight on respiratory metabolism of brook trout. He found that the rate of increase in scope with rising temperatures was progressively reduced for larger fish.

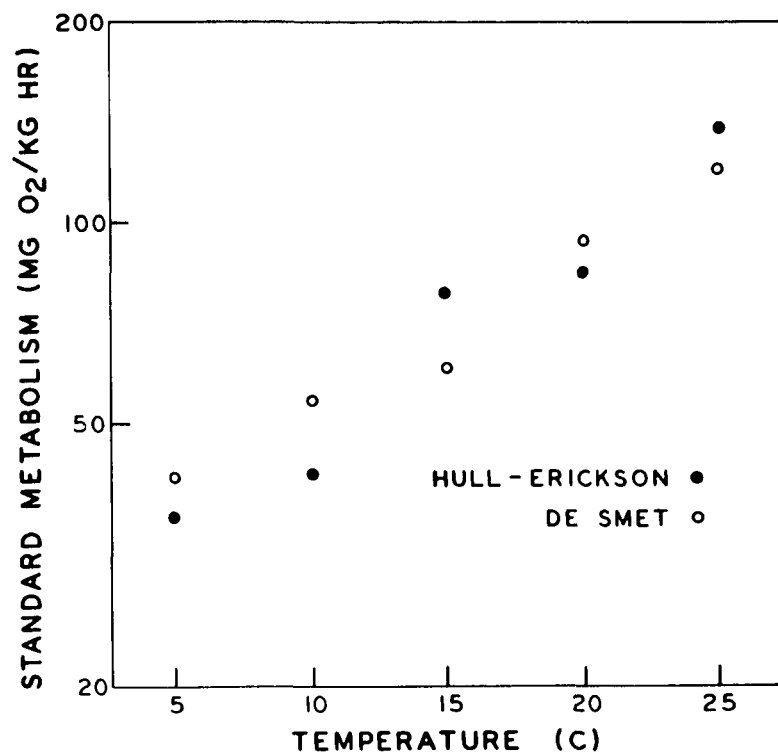


Figure 9. Temperature and standard metabolism, rainbow trout, Hull-Erickson and De Smet.

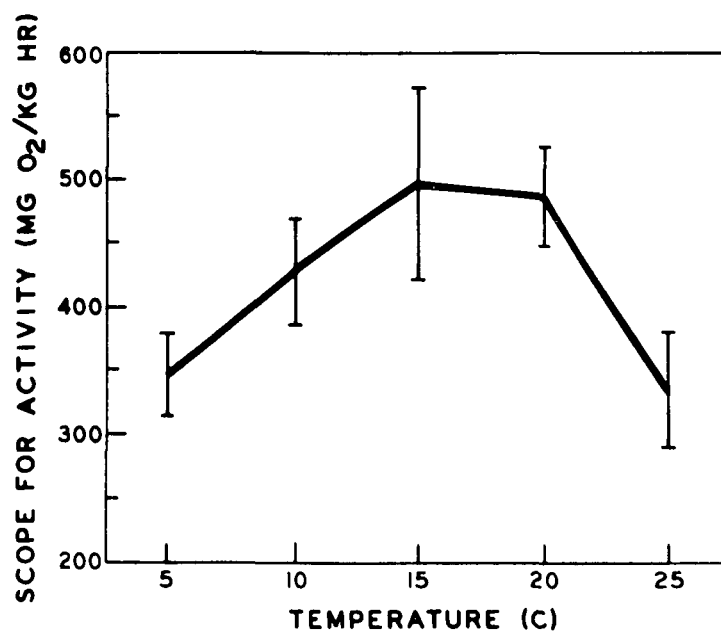


Figure 10. Temperature and scope for activity, rainbow trout, Hull-Erickson (The points represent the mean difference between active and standard metabolic rates ± 2 S.E.).

De Smet strain. Scope for activity increased with increasing temperature to 20 C (Figure 11). Maximum scope was 498 ± 22 mg/kg hr at 20 C and decreased slightly to 492 ± 37 and 468 ± 26 mg/kg hr at 15 and 25 C, respectively. Minimum scope was 324 ± 16 mg/kg hr at 5 C and increased to 438 ± 20 mg/kg hr at 10 C.

Scope for activity at 5 C was significantly less than at 10 and 15 C ($\underline{z} > 4.38$). Scope for activity at 10 C did not differ significantly from scope at 15 and 25 C ($\underline{z} < 1.27$), but was significantly less than scope at 20 C ($\underline{z} = 2.03$). Scope was not significantly different at 15 and 20 C, or 20 and 25 C ($\underline{z} < 0.90$). Significant differences in mean weights of fish at 5, 10, and 25 C, and also between mean fish weights at 15 and 25 C prevented statistical comparison between these temperatures. In each case, fish acclimated to the lower temperatures weighed significantly more than those at higher temperatures. In accordance with Job's findings (1955), an increase in mean weights of fish acclimated to higher temperatures would tend to further decrease scope for activity at the higher temperatures. Therefore, these values are probably maximal for scope at the higher temperatures.

Hull-Erickson vs De Smet strain. Scopes for activity of hatchery and wild trout at different temperatures are compared in Figure 12. Scope for activity values for the two strains were not significantly different at 5, 10, 15, and 20 C ($\underline{z} < 1.09$). Wild trout, however, had a significantly higher scope at 25 C than hatchery trout ($\underline{z} = 3.88$). This difference in scope at 25 C was caused by the effect of temperature on active

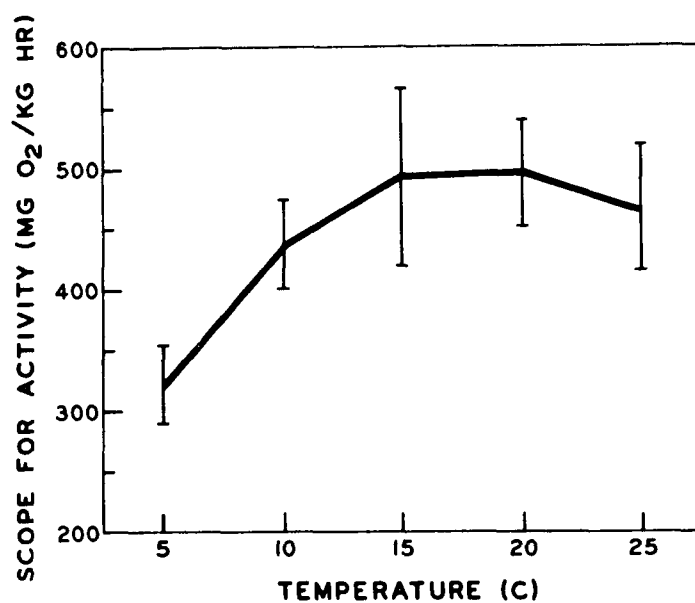


Figure 11. Temperature and scope for activity, rainbow trout, De Smet (The points represent the mean difference between active and standard metabolic rates ± 2 S.E.).

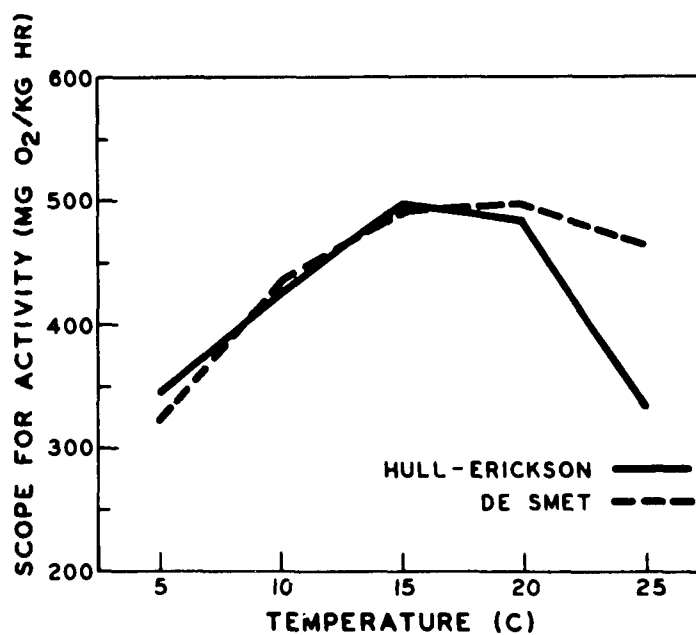


Figure 12. Temperature and scope for activity, rainbow trout, Hull-Erickson and De Smet.

metabolism of the two strains, since standard metabolic rates for the two strains at this temperature were not significantly different.

Starvation and Respiratory Metabolism of Hull-Erickson Rainbow Trout

Active metabolism

Active metabolic rates of 50 fish acclimated to 15 C and starved for 1, 2, 3, 6, and 9 days were determined during July 6-25, 1967 (Table 13). Fish weighed 146.1 to 328.4 g and were 25.0 to 29.5 cm long. With the exception of fish starved for 9 days and exercised at 67 to 91 cm/sec velocities (2.5-3.6 L/sec), all fish were forced to swim at 91 cm/sec velocity (3.0-4.0 L/sec).

Starvation had no significant effect on rates of active metabolism (Figure 13). Mean active rates of oxygen consumption following 1 and 2 days starvation remained unchanged and were 510 ± 12 and 510 ± 21 mg/kg hr, respectively. Active metabolic rates were relatively unaffected after 3 days starvation but increased slightly following 6 days starvation to 558 ± 16 mg/kg hr. A decrease in active metabolism to 528 ± 17 mg/kg hr occurred following 9 days starvation. Active metabolic rates at the 5 starvation levels were not significantly different ($F=1.83$ with 4 and 44 d.f.).

Standard metabolism

Standard metabolic rates were determined June 28 through August 14, 1967, for 45 trout acclimated to 15 C and starved for the aforesaid

Table 13. Active metabolism of starved Hull-Erickson trout acclimated to 15 C

Starvation period		Fork length (cm)	Wet weight (g)	Sex		Active metabolism (mg/kg hr)	Swimming speed	
				M	F		(cm/sec)	(L/sec) ^a
1 day	Number	(10)	(10)	5	5	(10)	(10)	(10)
	Range	23.0-29.0	163.4-328.4	-	-	408-612	91	3.1-4.0
	Mean	26.0	221.6	-	-	510	-	3.6
	S.D.	2.1	48.6	-	-	66	-	1.0
	S.E.	0.7	15.4	-	-	21	-	0.3
2 days	Number	(10)	(10)	4	6	(10)	(10)	(10)
	Range	25.5-28.0	182.2-258.5	-	-	456-558	91	3.3-3.6
	Mean	26.6	224.5	-	-	510	-	3.5
	S.D.	1.0	27.5	-	-	37	-	0.1
	S.E.	0.3	8.7	-	-	12	-	0.0
3 days	Number	(10)	(10)	5	5	(10)	(10)	(10)
	Range	23.0-29.0	146.1-266.3	-	-	372-594	91	3.0-4.0
	Mean	27.0	226.4	-	-	498	-	3.5
	S.D.	2.5	43.8	-	-	72	-	0.4
	S.E.	0.8	13.9	-	-	23	-	0.1

Table 13. Continued

Starvation period		Fork length (cm)	Wet weight (g)	Sex		Active metabolism (mg/kg hr)	Swimming speed	
				M	F		(cm/sec)	(L/sec) ^a
6 days	Number	(10)	(10)	5	5	(10)	(10)	(10)
	Range	25.0-29.5	190.1-270.1	-	-	504-660	91	3.1-3.6
	Mean	26.9	226.6	-	-	558	-	3.4
	S.D.	1.3	31.2	-	-	50	-	0.1
	S.E.	0.4	9.9	-	-	16	-	*
9 days	Number	(10)	(10)	3	7	(10)	(10)	(10)
	Range	25.5-27.5	186.5-246.2	-	-	468-654	67-91	2.5-3.6
	Mean	26.2	212.6	-	-	528	83	3.1
	S.D.	0.7	19.2	-	-	55	11	0.8
	S.E.	0.2	6.1	-	-	17	3	0.3

* < 0.05

^a Fish body lengths per sec.

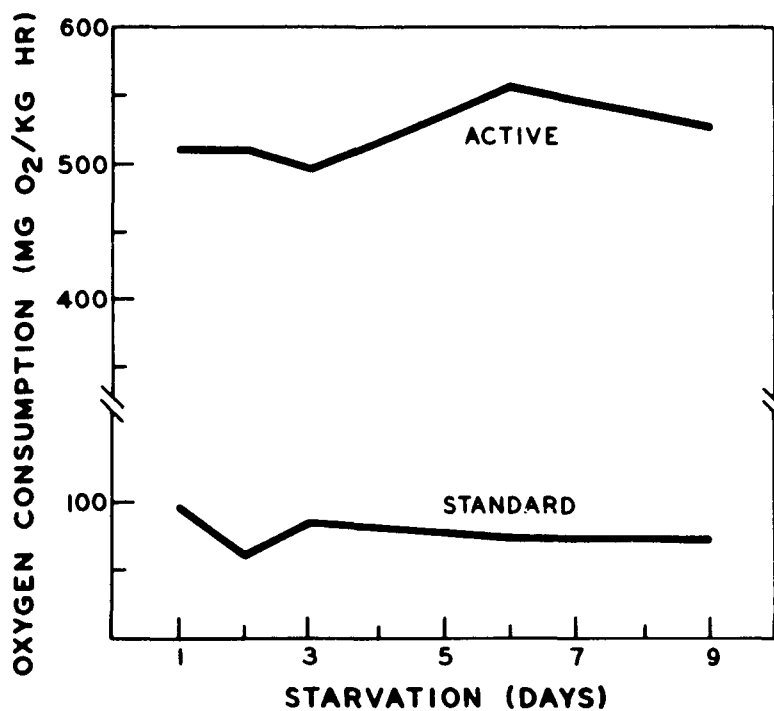


Figure 13. Active and standard metabolism, rainbow trout, Hull-Erickson, 15 C, starved 1, 2, 3, 6, and 9 days.

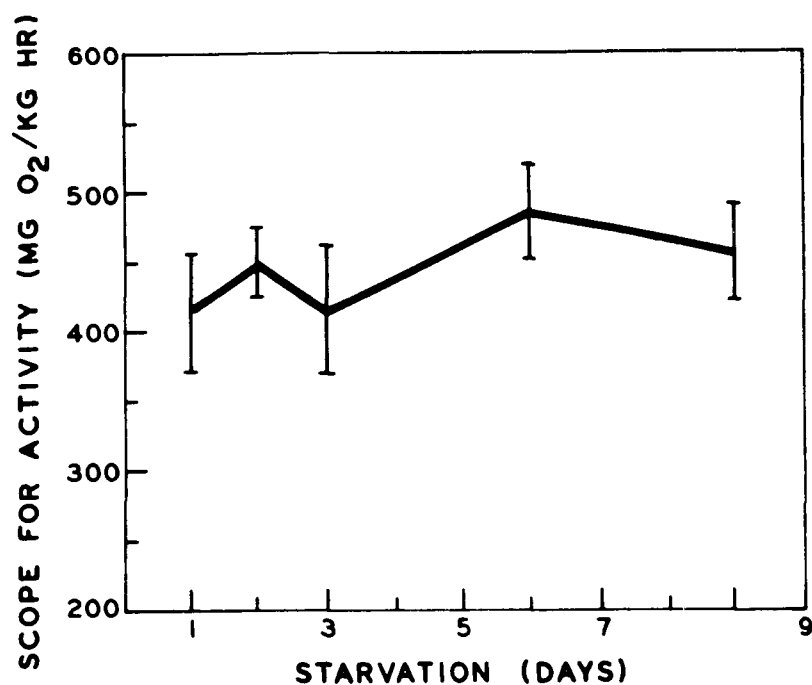


Figure 14. Scope for activity, rainbow trout, Hull-Erickson, 15 C, starved 1, 2, 3, 6, and 9 days (The points represent the mean difference between active and standard metabolic rates ± 2 S.E.).

periods (Table 14). Fish weighed 149.5 to 311.0 g and were 23.5 to 30.5 cm long. Standard metabolism decreased significantly after 2 days starvation but increased slightly with continued starvation (Figure 13). Maximum standard oxygen consumption was 96 ± 5 mg/kg hr following 1 day starvation and decreased to a minimum of 60 ± 7 mg/kg hr after starvation for 2 days. Standard metabolism increased to 84 ± 7 mg/kg hr in 3 days starvation, decreased to 72 ± 7 mg/kg hr in 6 days, and was unchanged at 9 days starvation.

Starvation had a significant effect on standard oxygen consumption ($F=3.42$ with 4 and 39 d.f.). Standard metabolism differed significantly at 1 and 2 days of starvation. No other significant differences occurred (Table 15).

Scope for activity

Scope for activity increased with starvation up to 6 days (Figure 14). Scope increased from a minimum of 414 ± 22 mg/kg hr at 1 day starvation to a maximum of 486 ± 17 mg/kg hr after 6 days starvation. Scope decreased at 9 days starvation to 456 ± 18 mg/kg hr. Scope for activity values following 2 and 3 days starvation were 450 ± 14 and 414 ± 24 mg/kg hr, respectively,

Scope for activity at 6 days starvation differed significantly from scope at 1 and 3 days starvation ($z \geq 2.42$). Scope for activity was not significantly different at other starvation levels ($z \leq 1.61$).

Table 14. Standard metabolism of starved Hull-Erickson trout acclimated to 15 C

Starvation period		Fork length (cm)	Wet weight (g)	Sex		Standard metabolism (mg/kg hr)
				M	F	
1 day	Number	(9)	(9)	4	5	(9)
	Range	24.5-30.5	152.9-299.5	-	-	78-120
	Mean	27.3	233.4	-	-	96
	S.D.	2.0	40.5	-	-	15
	S.E.	0.7	13.5	-	-	5
2 days	Number	(9)	(9)	2	7	(9)
	Range	24.5-30.0	169.0-289.3	-	-	24-90
	Mean	27.4	228.8	-	-	60
	S.D.	1.8	39.0	-	-	20
	S.E.	0.6	13.0	-	-	7
3 days	Number	(9)	(9)	5	4	(9)
	Range	25.0-29.5	178.0-311.0	-	-	48-126
	Mean	26.7	228.5	-	-	84
	S.D.	1.3	40.7	-	-	22
	S.E.	0.4	13.6	-	-	7

Table 14. Continued

Starvation period		Fork length (cm)	Wet weight (g)	Sex		Standard metabolism (mg/kg hr)
				M	F	
6 days	Number	(9)	(9)	5	4	(9)
	Range	24.5-29.5	181.9-266.0	-	-	48-102
	Mean	27.0	224.3	-	-	72
	S.D.	1.6	31.4	-	-	20
	S.E.	0.5	10.5	-	-	7
9 days	Number	(9)	(9)	4	5	(9)
	Range	23.5-28.5	149.5-245.0	-	-	54-102
	Mean	26.6	213.5	-	-	72
	S.D.	1.7	31.1	-	-	14
	S.E.	0.6	10.4	-	-	5

Table 15. Standard metabolic rates of Hull-Erickson trout following different periods of starvation--expressed as mg/kg hr (Values joined by underscore are not significantly different from each other at the 5 per cent level, Newman-Keul's multiple range test)

	Days of starvation				
	2	3	6	9	1
No. of observations	(9)	(9)	(9)	(9)	(9)
Mean standard rates	60	84	72	<u>72</u>	96

Time of Day and Respiratory Metabolism
of De Smet Rainbow Trout

Measurements of active and standard oxygen consumption were restricted to two periods during the daylight hours--the forenoon (7:00 AM to 1:00 PM) and the afternoon (1:00 PM to 7:00 PM). Fish were acclimated to 15 C and subjected to continuous light, and controlled periods of light and dark coinciding with the natural photoperiod. Experiments were conducted December 4-29, 1967.

Active metabolism

Controlled light and dark. Active metabolic rates were determined for 12 and 11 fish in the forenoon and afternoon periods, respectively (Table 16). The former weighed 153.4 to 261.3 g, were 24.0 to 29.2 cm long, and were exercised at 73 to 107 cm/sec velocities (3.1-4.3 L/sec). The latter weighed 88.0 to 247.0 g, were 19.5 to 32.8 cm long, and swam

Table 16. Time of day and active metabolism of De Smet trout acclimated to 15 C and subjected to controlled periods of light and dark

Time of day		Fork length (cm)	Wet weight (g)	Sex		Active metabolism (mg/kg hr)	Swimming speed	
				M	F		(cm/sec)	(L/sec) ^a
7:00 AM to 1:00 PM	Number	(12)	(12)	6	6	(12)	(12)	(12)
	Range	24.0-29.2	153.4-261.3	-	-	456-666	73-107	3.1-4.3
	Mean	26.8	219.5	-	-	534	101	3.7
	S.D.	1.9	46.4	-	-	66	9	0.4
	S.E.	0.5	13.4	-	-	19	3	0.1
1:00 PM to 7:00 PM	Number	(11)	(11)	2	9	(11)	(11)	(11)
	Range	19.5-32.8	88.0-247.0	-	-	456-696	91-107	3.1-5.3
	Mean	24.9	187.1	-	-	558	98	4.0
	S.D.	4.0	96.3	-	-	78	6	0.7
	S.E.	1.2	29.0	-	-	24	2	0.2

^aFish body lengths per sec.

at velocities of 91 to 107 cm/sec (3.1-5.3 L/sec). Mean active metabolic rates of fish in the forenoon and afternoon were 534 ± 19 and 558 ± 24 mg/kg hr, respectively, and were not significantly different ($F=0.05$ with 1 and 20 d.f.).

Continuous light. Mean active metabolic rates of 12 fish processed during each of the forenoon and afternoon periods are given in Table 17. Fish used in the forenoon experiments weighed 86.3 to 324.8 g and were 19.0 to 30.6 cm long. Fish in afternoon experiments weighed 100.0 to 575.9 g and were 21.0 to 36.0 cm long. Fish were exercised at 91 to 107 cm/sec velocities (2.5-4.8 L/sec). Mean active rates of oxygen consumption were 528 ± 24 mg/kg hr in the morning to 534 ± 23 mg/kg hr in the afternoon but were not significantly different ($F=0.74$ with 1 and 21 d.f.).

Standard metabolism

Controlled light and dark. Twelve measurements of standard oxygen consumption were made during each of the forenoon and afternoon periods (Table 18). Fish processed in the forenoon weighed 109.7 to 305.9 g and were 23.0 to 29.7 cm long. Those used during afternoon experiments weighed 85.6 to 308.9 g and were 20.0 to 29.6 cm long. Mean standard rates of oxygen uptake decreased from 84 ± 7 mg/kg hr in the forenoon to 78 ± 8 mg/kg hr in the afternoon, but were not significantly different ($F=0.15$ with 1 and 19 d.f.).

Continuous light. Results of standard metabolism experiments involving 12 measurements of fish during each of the forenoon and afternoon

Table 17. Time of day and active metabolism of De Smet trout acclimated to 15 C and subjected to continuous light

Time of day		Fork length (cm)	Wet weight (g)	Sex		Active metabolism (mg/kg hr)	Swimming speed	
				M	F		(cm/sec)	(L/sec) ^a
7:00 AM to 1:00 PM	Number	(12)	(12)	2	10	(12)	(12)	(12)
	Range	19.0-30.6	86.3-324.8	-	-	432-720	91-104	3.0-4.8
	Mean	26.1	214.4	-	-	528	101	3.9
	S.D.	3.4	73.3	-	-	84	6	0.3
	S.E.	1.0	21.2	-	-	24	2	0.1
1:00 PM to 7:00 PM	Number	(12)	(12)	2	10	(12)	(12)	(12)
	Range	21.0-36.0	100.0-575.9	-	-	408-684	91-107	2.5-4.4
	Mean	27.8	256.5	-	-	534	101	3.6
	S.D.	3.9	127.2	-	-	78	21	0.5
	S.E.	1.1	36.7	-	-	23	6	0.1

^aFish body lengths per sec.

Table 18. Time of day and standard metabolism of De Smet trout acclimated to 15 C and subjected to controlled periods of light and dark

Time of day		Fork length (cm)	Wet weight (g)	Sex		Standard metabolism (mg/kg hr)
				M	F	
7:00 AM to 1:00 PM	Number	(11)	(11)	6	5	(11)
	Range	23.0-29.7	109.7-305.9	-	-	48-120
	Mean	26.2	209.9	-	-	84
	S.D.	1.9	51.6	-	-	23
	S.E.	0.6	15.6	-	-	7
1:00 PM to 7:00 PM	Number	(11)	(11)	6	5	(11)
	Range	20.0-29.6	85.6-308.9	-	-	48-132
	Mean	26.8	237.5	-	-	78
	S.D.	2.8	68.6	-	-	25
	S.E.	0.8	20.7	-	-	8

periods are given in Table 19. Fish used in the morning experiments weighed 105.3 to 380.5 g and were 21.7 to 31.0 cm long. Afternoon measurements were made using fish that weighed 115.8 to 330.4 g and were 22.0 to 30.5 cm long. Mean rates of standard metabolism were 72 ± 6 mg/kg hr in the forenoon and 72 ± 9 mg/kg hr in the afternoon.

Scope for activity

Scope for fish maintained under controlled conditions of light and dark increased from 450 ± 20 mg/kg hr in the forenoon to 480 ± 25 mg/kg hr in the afternoon. Scope for activity of fish maintained under constant light was 456 ± 25 mg/kg hr in the forenoon and increased to 462 ± 25 mg/kg hr in the afternoon. No differences were statistically significant. Time of day and light conditions, therefore, had no influence on scope for activity.

Season and Active Metabolism of Hull-Erickson Rainbow Trout

Measurements of active metabolism of hatchery trout acclimated to 15 C were made on five separate occasions between October, 1966 to August, 1967. Active metabolism was determined in October–November, 1966, and February, April, July, and August, 1967. This provided an opportunity to investigate seasonal influence on active rates of oxygen consumption. Ten to 49 measurements were made at any one time. Active metabolism measurements of male and female fish were separated to permit investigation of seasonal effects on the sexes.

Table 19. Time of day and standard metabolism of De Smet trout acclimated to 15 C and subjected to continuous light

Time of day		Fork length (cm)	Wet weight (g)	Sex		Standard metabolism (mg/kg hr)
				M	F	
7:00 AM to 1:00 PM	Number	(12)	(12)	7	5	(12)
	Range	21.7-31.0	105.3-380.5	-	-	36-108
	Mean	27.1	232.4	-	-	72
	S.D.	3.2	85.5	-	-	22
	S.E.	0.9	24.7	-	-	6
1:00 PM to 7:00 PM	Number	(12)	(12)	4	8	(12)
	Range	22.0-30.5	115.8-330.4	-	-	24-126
	Mean	26.3	209.9	-	-	72
	S.D.	2.4	55.4	-	-	30
	S.E.	0.7	16	-	-	9

Males

Mean active metabolic rates of males determined at five different times during 11 consecutive months are given in Table 20. Fish weighed 95.8 to 339.8 g and were 19.2 to 30.3 cm long. Fish swam at velocities of 67 to 91 cm/sec (2.7-4.4 L/sec). Active rate of oxygen consumption was 546 ± 28 mg/kg hr during October-November, 1966, and increased to a maximum of 618 ± 22 mg/kg hr in February, 1967 (Figure 15). Active rates remained relatively high in April, but decreased to a minimum of 522 ± 38 mg/kg hr in July. Mean active metabolic rates of males at the five test periods were not significantly different ($F=1.38$ with 4 and 32 d.f.).

Females

Seasonal trends in active metabolism for females was similar to that for males (Figure 15). Fish weighed 91.6 to 341.6 g and were 20.2 to 30.5 cm long. Fish swam at 67 to 104 cm/sec velocities (2.6-4.5 L/sec). Mean active rates of oxygen uptake increased from a minimum of 492 ± 10 mg/kg hr in October-November, 1966 to a maximum of 600 ± 29 mg/kg hr in February, 1967 (Table 21).

The effect of season upon active rates of oxygen consumption of females was significant ($F=3.18$ with 4 and 52 d.f.). Active metabolic rates in October-November were significantly lower than those in February and April (Table 22).

Table 20. Effect of season on active metabolism of male Hull-Erickson trout acclimated to 15 C

Season		Fork length (cm)	Wet weight (g)	Active metabolism (mg/kg hr)	Swimming speed (cm/sec) (L/sec) ^a	
October 26 to November 5 1966	Number	(15)	(15)	(15)	(15)	(15)
	Range	21.6-30.3	124.3-339.8	384-738	91	3.0-4.2
	Mean	25.9	227.2	546	-	3.5
	S.D.	2.7	69.2	108	-	0.4
	S.E.	0.7	17.9	28	-	0.1
February 1 to February 4 1967	Number	(9)	(9)	(9)	(9)	(9)
	Range	19.2-26.5	95.8-188.7	522-702	67-91	2.7-4.4
	Mean	23.6	143.7	618	84	3.6
	S.D.	2.5	38.5	66	9	0.6
	S.E.	0.8	12.8	22	3	0.2
April 22 to April 26 1967	Number	(4)	(4)	(4)	(4)	(4)
	Range	25.4-26.2	190.0-206.5	510-636	91	3.5-3.6
	Mean	25.8	200.5	588	-	3.6
	S.D.	0.4	7.5	60	-	0.1
	S.E.	0.2	3.8	30	-	0.1

Table 20. Continued

Season		Fork length (cm)	Wet weight (g)	Active metabolism (mg/kg hr)	Swimming speed (cm/sec) (L/sec) ^a	
July 6 1967	Number	(5)	(5)	(5)	(5)	(5)
	Range	23.0-28.5	174.5-256.8	408-612	91	3.2-4.0
	Mean	26.1	212.2	522	-	3.5
	S.D.	2.3	32.7	84	-	0.3
	S.E.	1.0	14.6	38	-	0.1
August 15 to August 19 1967	Number	(5)	(5)	(5)	(5)	(5)
	Range	27.0-28.5	208.0-283.0	534-612	76	2.7-2.8
	Mean	27.4	231.1	558	-	2.8
	S.D.	0.7	30.0	31	-	*
	S.E.	0.3	13.4	14	-	*

* <0.05

^aFish body lengths per sec.

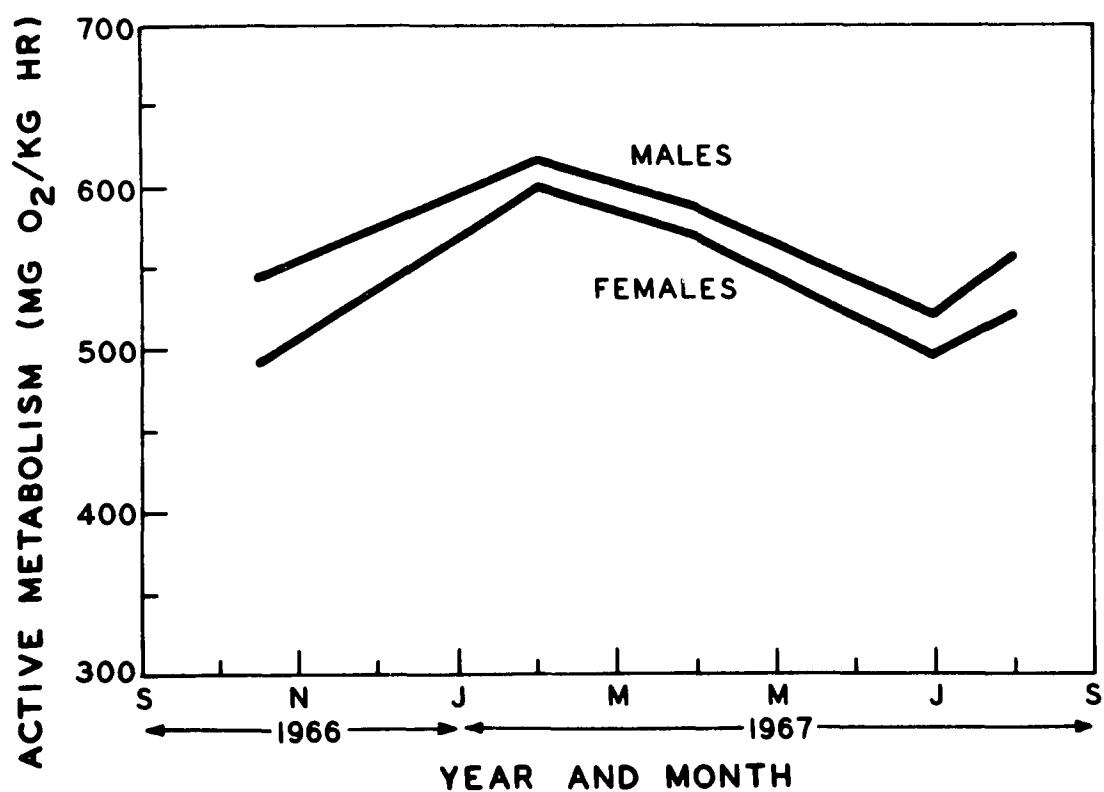


Figure 15. Season and active metabolism, rainbow trout, Hull-Erickson, 15 C.

Table 21. Effect of season on active metabolism of female Hull-Erickson trout acclimated to 15 C

Season		Fork length (cm)	Wet weight (g)	Active metabolism (mg/kg hr)	Swimming speed (cm/sec) (L/sec) ^a	
	Number	(29)	(29)	(20)	(29)	(29)
October 26	Range	20.2-30.5	97.3-341.6	402-600	91	3.1-4.5
to	Mean	25.4	203.1	492	-	3.6
November 5	S.D.	2.3	55.6	54	-	0.3
1966	S.E.	0.4	10.3	10	-	0.1
	Number	(14)	(14)	(14)	(14)	(14)
February 1	Range	20.7-27.5	91.6-199.5	360-816	67-104	2.6-4.4
to	Mean	23.4	142.5	600	88	3.8
February 4	S.D.	2.5	42.1	108	12	0.5
1967	S.E.	0.7	11.3	29	3	0.1
	Number	(6)	(6)	(6)	(6)	(6)
April 22	Range	25.2-27.0	187.0-240.9	468-672	91	3.4-3.6
to	Mean	26.1	215.6	570	-	3.5
April 26	S.D.	0.7	18.6	78	-	0.1
1967	S.E.	0.3	7.6	32	-	*

Table 21. Continued

Season		Fork length (cm)	Wet weight (g)	Active metabolism (mg/kg hr)	Swimming speed (cm/sec) (L/sec) ^a	
July 6 1967	Number	(5)	(5)	(5)	(5)	(5)
	Range	23.5-29.0	163.4-328.4	438-558	91	3.1-3.9
	Mean	26.0	231.0	498	-	3.5
	S.D.	2.2	63.5	53	-	0.1
	S.E.	1.0	28.4	24	-	*
August 15 to August 19 1967	Number	(5)	(5)	(5)	(5)	(5)
	Range	27.5-30.0	223.0-259.3	336-654	76	2.6-2.8
	Mean	28.5	244.1	528	-	2.7
	S.D.	1.0	15.3	64	-	0.1
	S.E.	0.4	6.8	29	-	*

* <0.05

^a Fish body lengths per sec.

Table 22. Effect of season on active metabolic rates of female Hull-Erickson trout acclimated to 15 C--expressed as mg/kg hr (Values joined by underscore and dotted line are not significantly different from each other at the 5 per cent level, Newman-Keul's range test)

	Season				
	1967		1966	1967	
	Feb.	April	Oct.-Nov.	July	Aug.
No. of observations	(14)	(6)	(29)	(5)	(24)
Mean active rates	<u>600</u>	<u>570</u>	<u>492</u>	<u>498</u>	<u>528</u>

Sex and Active Metabolism of
Hull-Erickson Rainbow Trout

Separation of the sexes to determine seasonal effects on active metabolism permitted further comparisons for sexual differences in active rates of oxygen consumption (Figure 15). Since the mean weights of males and females were not significantly different, it was possible to compare the mean rates of active metabolism of males and females acclimated to 15 C at each season. In all five cases, active metabolic rates of males were greater than those of females. The relative difference in all cases was 3-10 per cent of the female rate. Active rates of males were significantly higher at the 10 per cent level than those of females during October-November, 1966, and August, 1967 ($\underline{z}=1.82$ and 1.71). Highly significant differences were found, however, when the active rates determined during October-November, 1966 were treated by analysis of

covariance. The limited number of measurements made during the other seasonal periods precluded covariance analyses in these cases. In October–November, active rates of males were significantly higher than those of females at the 1 per cent level ($F=24.1$ with 1 and 41 d.f.).

Active metabolic rates determined during October–November, 1966 for fish acclimated to 10 C were also analyzed by covariance (Table 23). Males had significantly higher active rates than females at the 6 per cent level ($F=4.11$ with 1 and 34 d.f.).

Table 23. Effect of sex on active metabolism of Hull-Erickson trout acclimated to 10 C

Sex		Fork length (cm)	Wet weight (g)	Active metabolism (mg/kg hr)	Swimming speed	
					(cm/sec)	(L/sec) ^a
Males	Number	(23)	(23)	(23)	(23)	(23)
	Range	21.0-28.8	105.6-321.2	336-606	91	3.1-4.4
	Mean	25.1	213.9	462	-	3.7
	S.D.	2.4	66.4	72	-	0.4
	S.E.	0.5	13.8	15	-	0.1
Females	Number	(14)	(14)	(14)	(14)	(14)
	Range	23.0-29.7	143.0-353.8	300-510	91	3.1-4.0
	Mean	26.4	234.2	408	-	3.5
	S.D.	2.3	66.3	66	-	0.3
	S.E.	0.6	17.7	18	-	0.1

^aFish body lengths per sec.

DISCUSSION

Suitability of the Apparatus

Active respirometer

In this study of rainbow trout metabolism, estimates of active metabolic rates were made without precise control of swimming speed. Though swimming speed was not precisely controlled, the type of respirometer employed was well suited to estimate active metabolic rates. Water velocities as high as 107 cm/sec (5.1 L/sec) could be generated with this apparatus, although fish were usually exercised at a velocity of about 91 cm/sec (3.0-4.5 L/sec) during "forced short term" periods. Preliminary studies indicated that a velocity of this magnitude was sufficient to induce the fish to perform at maximum rate of oxygen uptake. This forced exercise, and probably in some cases, the additional stimulus of electric shock and/or excitement was sufficient for full expression of the active metabolic rate. Generally, measurements of active rates of oxygen consumption were relatively unaffected by such behavioral characteristics as excitability, restlessness, and spontaneous movement that contribute to the variability of estimates of standard metabolism. This is to be expected, however, as small changes in fish behavior have a proportionately greater effect on standard metabolism than on active metabolism (Brett, 1965). This contention was supported in the present study by the small amount of variation observed in active metabolism

Standard respirometer

The type of flowmeter or thermostatic heater probe used to measure spontaneous activity in this study was used by McKenzie (1960), Beamish and Mookherjee (1964), and more recently by Mather and Shrivastava (1967). These investigators, using relatively small respirometers, found a linear arithmetic relationship between oxygen consumption and spontaneous activity over the range of activities observed.

In the present study, however, this arithmetic relationship was not always linear over the range of activities encountered. Presumably, the considerably larger respirometers used in this study permitted increased swimming activity. In addition, some experiments involved standard oxygen consumption measurements of "excited" or "nervous" fish that exhibited high levels of spontaneous activity. Prolonged habituation of the fish to the chamber, exclusion of light from the chamber, and other precautions taken to reduce the fish to a quiescent state were unsuccessful. For these individuals, the relation of oxygen consumption to activity was a concave curvilinear arithmetic plot. This curvilinear function was converted to a linear relation by plotting the logarithm of oxygen consumption against activity. Therefore, rather than discard results obtained from experiments involving the excited fish, all estimates were obtained by a semi-logarithmic analyses of oxygen consumption and activity. In the majority of experiments where spontaneous activity was low, the linear arithmetic relationship was relatively unaffected by the semi-logarithmic analyses other than to increase the value of the "a" intercept,

and consequently, increase the estimate for standard metabolism.

Standard metabolism measurements of rainbow trout were subject to considerable variation. This was not unusual, however, because of the "unavoidably high experimental error" usually associated with estimates of standard oxygen consumption (Beamish and Dickie, 1967, p. 220). Smit (1965) and Brett (1964, 1965) found a semi-logarithmic relationship between oxygen consumption and swimming speeds, and they extrapolated this relation to zero activity to estimate standard metabolism. They discussed how estimates of oxygen consumption obtained in this manner can be affected by excitement, restlessness, and other types of fish behavior. The comparatively higher rates of oxygen consumption of fish spontaneously active at low levels of activity is attributed to the high cost of intermittent swimming which becomes more efficient at the faster more continuously maintained speeds (Beamish and Dickie, 1967).

Temperature, Active and Standard Metabolism, and Scope for Activity of Rainbow Trout

Active metabolism

The increased active metabolic rates of wild and hatchery trout with increased temperature to 20 and 15 C, respectively, followed by a decrease in active metabolism with further increase in temperature is typical for most species. However, the markedly different response of the two strains to 25 C suggests that wild trout may be more active in warm water. This is not surprising, since the increased scope for activity

associated with the higher active rates of the wild trout indicates more "available energy" for swimming (Brett, 1964, p. 1207).

Although active metabolism of both strains decreased at 25 C, active metabolic rates of wild trout at 25 C were not significantly different from those at 15 and 20 C. Similarly, Brett (1964) found no statistical difference between active metabolic rates of sockeye salmon acclimated to 15, 20, and 24 C, although the mean rates were observed to decrease at temperatures above 15 C. He attributed the decrease in active metabolism of sockeye salmon at high temperatures to the reduced oxygen available in solution at high temperatures, and "other factors of a physiological nature" (Brett, 1964, p. 1207). This could explain the decrease in active metabolic rates of hatchery and wild trout at the higher temperatures, but does not account for the difference in active metabolism between the two strains at 25 C.

Standard metabolism

Rainbow trout, like most fish species, have increased rates of standard oxygen consumption with increased temperature. The relationship between standard oxygen consumption and temperature for hatchery and wild strains of rainbow trout was linear when plotted on a semi-logarithmic grid. This relation differed somewhat from the convex curvilinear relationship frequently reported for other fish, particularly warm water species (Beamish and Mookherjee, 1964; Beamish, 1964c).

The linear semi-logarithmic relation described for rainbow trout

appears to be typical for cold water species. Beamish (1964c) described a similar relation for temperature and standard metabolism of the brook trout. Data presented by Gibson and Fry (1954) for the lake trout suggest a linear or possibly slightly concave semi-logarithmic relationship. Brett (1964) found a distinct concave semi-logarithmic relation between temperature and standard oxygen consumption of sockeye salmon.

Scope for activity

In the present study, an increase in scope for activity with increased temperature to 15 and 20 C was followed by a reduction in scope at higher temperature for hatchery and wild rainbow trout, respectively. This relationship of scope for activity to temperature has been observed for numerous fish species. Maximum scope for the rainbow trout occurred at temperatures higher than the 14 C preferred temperature reported for this species by Fry (1964). This indicated that maximum scope need not occur at the preferred temperature for the species.

The effect of temperature on active and standard metabolic rates of rainbow trout is similar to that reported by Brett (1964) for sockeye salmon. At temperatures up to 15 C, Brett found that active rates were 10-12 times the standard rate. Above 15 C the active-standard ratio decreased to about 4 times the standard rate at 24 C. The combined influence of temperature and activity elevated the respiratory metabolism by a factor of 22 times. In the present study, active rates of rainbow trout increased to 15 C and were approximately 9-11 times the standard rate. This ratio

decreased above 15 C to 3-5 times the standard rate at 25 C for hatchery and wild trout, respectively. The maximum active rate due to temperature and activity was about 16 times the lowest standard rate.

The difference in scope for activity of hatchery and wild trout acclimated to 25 C indicated a greater capacity for energy expenditure by the wild trout in waters of higher temperature. This additional energy available to wild trout may be an important survival mechanism during times of stress. Brett (1958, p. 80) indicated that scope for activity may be used as a means of "assessing environmental stress," if reduced scope indicates stress conditions. Wild trout, therefore, in water of high temperature may be better able to cope with temperature stress and have greater potential for survival under these conditions. This difference in scope could be attributed to the effects of hatchery and natural selection on the genetic composition of the two trout strains. Wild trout exposed to the temperature fluctuations and extremes encountered in nature may have retained a tolerance to high water temperatures.

Active and Standard Metabolism, and Scope for
Activity of Starved Hull-Erickson
Rainbow Trout

Active metabolism

The effect of starvation on active metabolism of fish has received little attention. Fry (1957, p. 37) reported results obtained by Barrett who investigated the effect of starvation on active metabolism of rainbow trout. Fish were maintained at approximately 11 C and were starved for

100 days. Barrett found "no essential difference" between the active rates of starved and control animals.

In the present study, similar results were obtained for rainbow trout starved from 1 to 9 days. No statistical difference was found between active rates determined for fish starved 1, 2, 3, 6, and 9 days. Results indicated that active metabolic rates of rainbow trout were independent of starvation during a 9-day period. This is understandable as starvation would probably not affect the total capacity of the fish to extract oxygen from the water, but may influence the metabolic pathways for oxygen utilization. Swimming fish forced to respire at their active rates, for example, would be expected to have less stamina immediately after feeding than starved fish. Presumably, the increased oxygen demand for food assimilation would decrease the oxygen supply available to the muscles which would reduce the time to muscle fatigue and decrease stamina. This contention is supported by findings of Thomas, Burrows, and Chenoweth (1964). They reported a decrease in stamina of fingerling salmon immediately after feeding followed by an increase in stamina to a maximum after 4 to 5 days starvation. More recently, however, Horak (1967) using wild and hatchery strains of rainbow trout reported high stamina immediately after feeding. Maximum stamina occurred at 3 days starvation but decreased with continued starvation. No attempt was made to explain these results.

Standard metabolism

A decrease in standard metabolism during the first 2 or 3 days of starvation has been reported by Beamish (1964a) to reflect a reduction in oxygen requirement for the assimilation of food. This would explain the significant reduction in standard metabolism of rainbow trout used in the present study following 2 days starvation.

Scope for activity

The increased scope for activity following starvation for 6 days reflects the influence of starvation on the standard metabolic rate. A reduction in the amount of oxygen required for general body maintenance (standard) would make more oxygen available for metabolic functions that demand maximum energy. Fish forced to swim at maximum sustained speeds may be less subject to oxygen debt following 6 days starvation than 1 day starvation. This might have survival value in fish-stocking situations where fish are planted in streams characterized by swift current.

Time of Day, Active and Standard Metabolism, and Scope for Activity of De Smet Rainbow Trout

Fry (1957) stated that fish may often show endogenous daily cycles of activity so that the level of metabolism may depend on the time of day in spite of maintenance of constant conditions. The purpose of the present experiment was to determine the effect of time of day on active and standard metabolic rates when these respiratory rates are not influenced by such

endogenous activity patterns. Active and standard metabolism were determined during morning and afternoon periods. In measurements of standard metabolism, spontaneous activity was quantified.

Considerable information is available concerning diurnal fluctuations in the respiratory metabolism of fish and there appears to be no general agreement concerning the existence of diurnal differences in the metabolic rate of fish (Winberg, 1956). Earlier studies like those of Clausen (1933, 1936), Oya and Kimata (1938), and Graham (1949), probably more correctly referred to as studies of routine metabolism by current definitions of respiratory rates of fish (Fry, 1967), described diurnal influence on fish metabolism by measuring rates of oxygen consumption throughout a 24-hour period. Although precautions were taken to reduce external stimuli and induce the fish to a quiescent state, the effect of spontaneous activity on respiratory metabolism was not determined.

In the present study of rainbow trout, active and standard rates of respiratory metabolism were not significantly affected by time of day when the possible influence of diurnal rhythms in spontaneous activity was eliminated. As standard and active metabolism of rainbow trout remained relatively unaltered during the daylight hours, it followed that scope for activity remained unchanged under these conditions. Higginbotham (1947) restricted his study of diurnal fluctuations in respiratory metabolism of channel catfish to the hours of daylight, and found higher rates of oxygen consumption in the afternoon than in the morning. He attributed this diurnal response to increased activity.

Winberg stated:

There is no basis whatever for postulating the existence of "rhythmic" fluctuations in metabolic rate that arise independently of environmental conditions. The diurnal changes in rate of oxygen consumption that are actually observed have their origin in the daily rhythm of light and dark and a complex of other natural factors that act not directly, but through changes in the pattern of the diurnal change in periods of rest and activity. (Winberg, 1956, p. 65)

Light

Active and standard metabolic rates of rainbow trout were unaffected by either controlled conditions of light and dark or continuous lighting.

References to metabolic rates of fish exposed to different lighting conditions are few. Pütter (1909) and Wells (1935) denied that light affects metabolic rate directly. Evans, Purdie, and Hickman (1962) have shown that routine metabolic rate of rainbow trout is unaffected by length of the photoperiod.

Season and Active Metabolism of Hull-Erickson Rainbow Trout

Active metabolism of rainbow trout fluctuated with season although not significantly in the majority of cases. Highest active metabolic rates of male and female rainbow trout occurred during the seasonal period associated with spawning (February). Reports in the literature concerning the influence of season on active metabolism are sparse. Wohlschlag and Juliano (1959) measured respiration rates of swimming bluegill and found respiration rates (undifferentiated) were higher in the winter than in the

spring, summer, and autumn seasons. They suggested that adaptation to cold temperature accounted for the higher metabolic rates during the winter months.

Seasonal effects on standard oxygen consumption have been more frequently reported (Wells, 1935; Privolnev, 1948; Beamish, 1964b). Increased respiration rates of fish species studied at various times of the year appear to be related to the reproductive cycle of each particular species. Beamish and Dickie (1967, p. 224) stated "seasonal changes in oxygen consumption are undoubtedly related to several factors of which photoperiod, temperature, and thyroid activity have received most consideration." Berg, Gorbman, and Kobayashi (1959) clearly demonstrated a relation between thyroxin production and season. They reported maximum thyroxin output during the spawning period of Fundulus heterclitus after which it decreased to a constant level.

Sex and Active Metabolism of Hull- Erickson Rainbow Trout

Mean active metabolic rates of males were consistently higher than females during all seasonal periods. Since the mean weights of males and females were not significantly different, mean active rates of the sexes were compared statistically. During October-November, 1966, and August, 1967, males had significantly higher rates than females ($\bar{z}=1.82$ and 1.71 ; $P<0.10$). The significant differences in active rate of oxygen consumption between the sexes was further supported at higher levels of

probability when active rates of oxygen consumption of males and females were analyzed by covariance analyses. At 15 C, active rates of males were significantly higher than females in October-November ($F=24.1$; $P<0.01$). Similarly, covariance analyses of active metabolism measurements of fish acclimated to 10 C showed significantly higher active rates of oxygen uptake by males than females ($F=4.11$; $P=0.06$). The consistently higher active rates of oxygen uptake observed for males and the statistically significant results obtained, particularly from covariance analyses, provide considerable evidence in support of the hypothesis that males have higher active metabolic rates than females.

Wohlschlag and Juliano (1959) found no significant difference in routine rates of male and female bluegill, although rates of oxygen consumption were consistently higher for females. They suggested that any effect of sex could better be established by confining experiments to more limited environmental fluctuations. Beamish (1964b) found that standard rates of oxygen uptake for male and female brook trout were approximately the same at different seasons. Similarly, he found no difference in standard rates between the sexes for brown trout, Salmo trutta, during different seasons, with the exception of the period September-November when standard rates of males were considerably higher than those of the females. He suggested that the males may have shown a seasonal response while females did not.

SUMMARY AND CONCLUSIONS

1. The oxygen consumption of rainbow trout, both a hatchery and a wild strain, the majority of which weighed 100 to 400 g, was measured after acclimation to 5, 10, 15, 20, and 25 C. Two measurements were made, vis., a standard rate obtained by extrapolation of the relation of the logarithm of oxygen consumption and spontaneous activity to zero activity, and an active rate determined by forcing fish to swim against high water velocity during short-term exercise periods. These two levels were taken as estimates of minimum and maximum rates of oxygen uptake, respectively. The influence of temperature on scope for activity (maximum-minimum) was determined.

2. Preliminary studies indicated that catchable-size rainbow trout would respire at active rates when exercised at approximately 91 cm/sec (3.0-4.5 L/sec). Temperatures of 5 and 25 C were observed to reduce swimming ability of rainbow trout, particularly the hatchery strain.

3. Active metabolism of hatchery trout increased from 384 ± 15 mg/kg hr at 5 C to a maximum of 576 ± 22 mg/kg hr at 15 C and decreased significantly to 474 ± 18 mg/kg hr at 25 C. Active rates of wild trout increased from 366 ± 16 mg/kg hr at 5 C to a maximum of 592 ± 19 mg/kg hr at 20 C, but were not significantly reduced at 25 C. At 25 C, active metabolic rates of wild trout were significantly higher than those of hatchery trout which indicated that wild trout may be more active in warm water.

4. Standard metabolism of hatchery and wild trout increased from 36 ± 2 and 42 ± 2 mg/kg hr at 5 C to 138 ± 12 and 120 ± 18 mg/kg hr at 25 C, respectively. A linear relationship was obtained when the logarithm of oxygen consumption was plotted against temperature. Standard metabolism of the two trout strains was significantly increased by approximately 3 to 4 times when the temperature was increased from 5 to 25 C ($F=17.65$ and 6.17 ; $P < 0.05$).

5. For hatchery and wild trout, respectively, scope for activity increased with temperature from 348 ± 16 and 324 ± 16 mg/kg hr at 5 C to a maximum of 498 ± 37 mg/kg hr at 15 C and 498 ± 22 mg/kg hr at 20 C. Minimum scope occurred at 5 C for wild trout, and at 25 C for hatchery trout (336 ± 23 mg/kg hr). Scope for activity values for the two trout strains were significantly different at 25 C only.

6. At temperatures up to 15 C active rates of hatchery and wild trout were about 9-11 times the standard rates. At 25 C, this ratio was reduced to 3 to 5 times the standard level. The highest active metabolic rate showed an approximately 16-fold increase over the lowest standard rate due to the combined influence of activity and temperature.

7. Active and standard metabolic rates were determined for hatchery trout acclimated to the maximum scope temperature (15 C) and starved for 1, 2, 3, 6, and 9 days. Starvation did not significantly affect active rates of oxygen uptake by hatchery rainbow trout.

8. Standard metabolism of hatchery trout was decreased significantly after starvation for 2 days, followed by a slight increase and

decrease to a constant level after 9 days starvation. The maximum standard rate of oxygen consumption was 96 ± 5 mg/kg hr after 1 day starvation and decreased to a minimum of 60 ± 7 mg/kg hr following starvation for 2 days ($F=3.42$; $P=0.05$).

9. Scope for activity of starved hatchery trout increased significantly from 414 ± 22 and 414 ± 24 mg/kg hr following 1 and 3 days starvation, respectively, to 486 ± 17 mg/kg hr after 6 days starvation. Starvation for 9 days resulted in a decrease in scope.

10. The effect of time of day on active and standard metabolism was determined for wild trout acclimated to 15 C. Active metabolic rates during morning and afternoon periods ranged from 528 ± 24 to 558 ± 24 mg/kg hr and were not significantly different. Similarly, no significant difference was found between forenoon and afternoon measurements of standard metabolism which varied from 72 ± 6 to 84 ± 7 mg/kg hr. Scope for activity remained unchanged from morning to afternoon.

11. Light conditions had no significant effect on active and standard metabolism of wild trout acclimated to 15 C and exposed to controlled conditions of light and dark or to continuous light.

12. Seasonal trends were indicated in active metabolic rates of males and females, although only seasonal rates of the latter were significantly affected. Maximum rates of active metabolism for both sexes occurred in February and were associated with the trout spawning season.

13. Males had consistently higher rates of active metabolism than females during all seasons. At 15 C, active metabolic rates of males

were significantly higher than those of females in August, 1967 ($\underline{z}=1.71$; $P < 0.10$), and also in October-November, 1966 ($F=24.1$; $P < 0.01$). Similarly, at 10 C, active rates of males were significantly higher than those of females in October-November, 1966 ($F=4.11$; $P=0.06$). This study indicated that male rainbow trout have higher active rates than females. However, further studies employing larger numbers of fish at all seasons are required to further substantiate these results.

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APPENDIX

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Table 24. Water velocity, temperature, and active metabolism of 60.1-146.6 g Hull-Erickson trout

Temperature (C)	Velocity (cm/sec)					
	43	67	73	76	91	104
Number ^a	(2)	(3)*	(3)	(2)**	(2)	
Weight range	109.3-115.3	106.3-125.4	109.2-137.0	-	109.4-126.4	
5 Mean weight	112.3	117.8	126.0	-	117.9	
Mean O ₂ uptake (mg/kg hr)	402	384	420	-	462	
Number		(2)		(3)*	(6)*	(2)
Weight range		115.0-119.0		127.0-135.1	109.0-146.6	137.0-139.1
10 Mean weight		117.0		129.7	124.6	138.1
Mean O ₂ uptake (mg/kg hr)		474		444	498	528
Number		(2)		(2)	(7)	
Weight range		106.1-106.5		95.8-102.3	91.6-131.3	
15 Mean weight		106.3		99.1	103.7	
Mean O ₂ uptake (mg/kg hr)		522		642	642	

Table 24. Continued

Temperature (C)	Velocity (cm/sec)					
	43	67	73	76	91	104
Number		(2)		(3)*	(5)	(4)**
Weight range		103.6-104.5		80.3-113.1	90.0-113.9	-
20 Mean weight		104.1		93.6	101.3	-
Mean O ₂ uptake (mg/kg hr)		618		540	642	-
Number	(2)	(5)*		(6)**	(7)	
Weight range	60.1-75.5	78.6-95.7		74.7-96.7	78.0-134.8	
25 Mean weight	67.8	83.1		86.3	90.5	
Mean O ₂ uptake (mg/kg hr)	372	618		714	708	

^a Number of fish exercised.

*Each asterisk represents a fatigued fish for which there was no measurement of oxygen consumption.

Table 25. Water velocity, temperature, and active metabolism of 134.6-241.4 g Hull-Erickson trout

	Temperature (C)	Velocity (cm/sec)						
		43	67	73	76	91	104	107
5	Number ^a	(2)	(2)	(2)	(2)	(3)**		
	Weight range	187.8-217.0	209.7-233.0	213.6-217.0	192.4-225.2	241.4		
	Mean weight	202.4	221.4	215.3	208.8	-		
	Mean O ₂ uptake (mg/kg hr)	390	498	396	342	348		
10	Number		(2)		(2)	(4)	(2)	(3)
	Weight range		167.0-212.1		201.0-218.9	210.8-229.8	204.3-215.6	192.6-217.4
	Mean weight		189.6		210.0	218.0	210.0	203.0
	Mean O ₂ uptake (mg/kg hr)		462		468	450	444	498
15	Number		(2)		(2)	(4)	(4)	
	Weight range		167.5-179.4		179.9-188.7	152.5-193.3	161.4-199.5	
	Mean weight		173.4		184.3	171.4	184.4	
	Mean O ₂ uptake (mg/kg hr)		558		528	651	642	
20	Number		(3)		(2)	(4)	(2)	(1)
	Weight range		134.6-181.3		158.2-181.2	169.3-197.7	178.0-186.6	193.8
	Mean weight		161.9		169.7	182.4	182.3	-
	Mean O ₂ uptake (mg/kg hr)		558		504	579	570	588
25	Number		(2)	(2)		(3)**		
	Weight range		153.3-168.5	150.7-155.9		141.2		
	Mean weight		160.9	153.3		-		
	Mean O ₂ uptake (mg/kg hr)		540	444		600		

^aNumber of fish exercised.

*Each asterisk represents a fatigued fish for which there was no measurement of oxygen consumption.

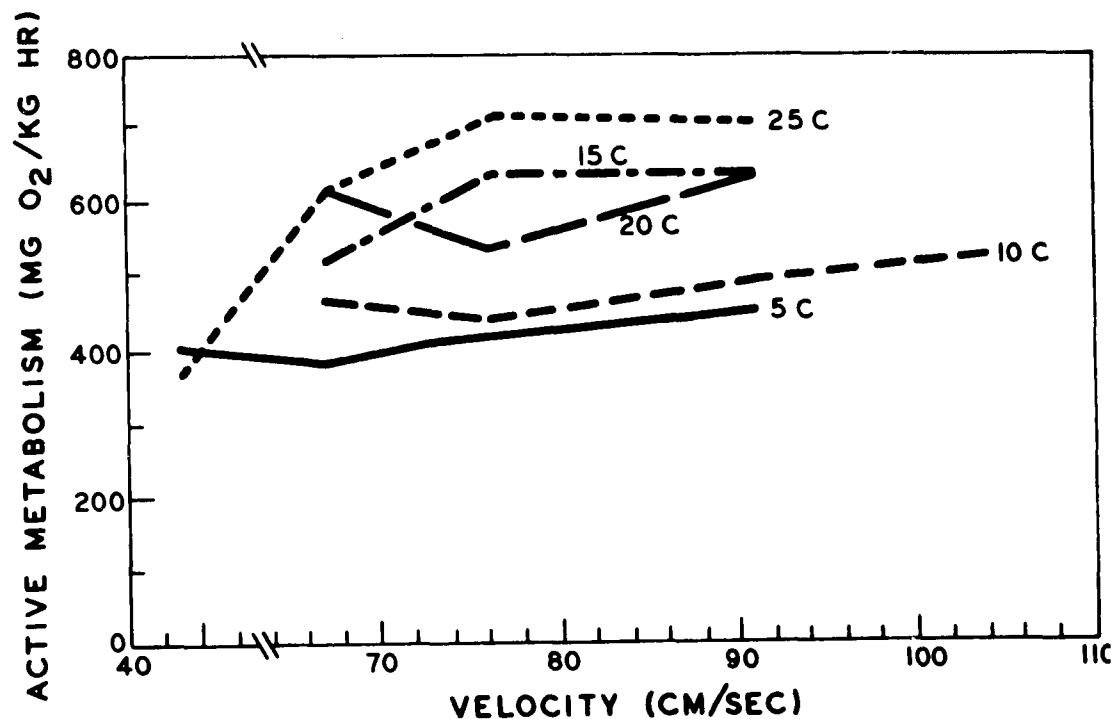


Figure 16. Water velocity, temperature, and active metabolism of 60.1-146.6 g Hull-Erickson trout.

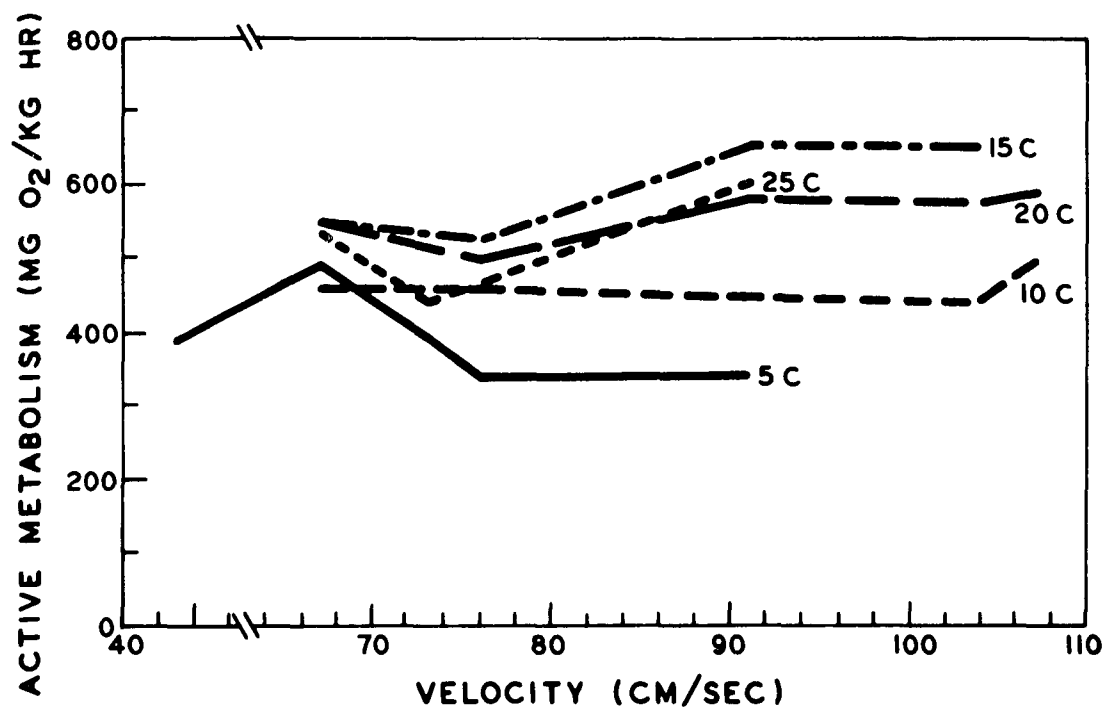


Figure 17. Water velocity, temperature, and active metabolism of 134.6-241.4 g Hull-Erickson trout.

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